Lecture 11:

Measuring Light:
Radiometry and Photometry

Computer Graphics and Imaging
UC Berkeley CS184/284A

Radiometry

Measurement system and units for illumination

Measure the spatial properties of light

- New terms: Radiant flux, intensity, irradiance, radiance

Perform lighting calculations in a physically correct manner

Assumption: geometric optics model of light

- Photons travel in straight lines, represented by rays

Radiometric & Photometric Terms & Units

Physics		Radiometry	Units	Photometry	Units
Energy	Q	Radiant Energy	Joules (W·sec)	Luminous Energy	Lumen·sec
Flux (Power)	Φ	Radiant Power	W	Luminous Power	Lumen (Candela sr)
Angular Flux Density	I	Radiant Intensity	W/sr	Luminous Intensity	Candela (Lumen/sr)
Spatial Flux Density	E	Irradiance (in) Radiosity (out)	W/m²	Illuminance (in) Luminosity (out)	Lux (Lumen/m²)
Spatio-Angular Flux Density	L	Radiance	W/m²/sr	Luminance	Nit (Candela/m²)

"Thus one nit is one lux per steradian is one candela per square meter is one lumen per square meter per steradian. Got it?" — *James Kajiya*

Light

Visible electromagnetic spectrum

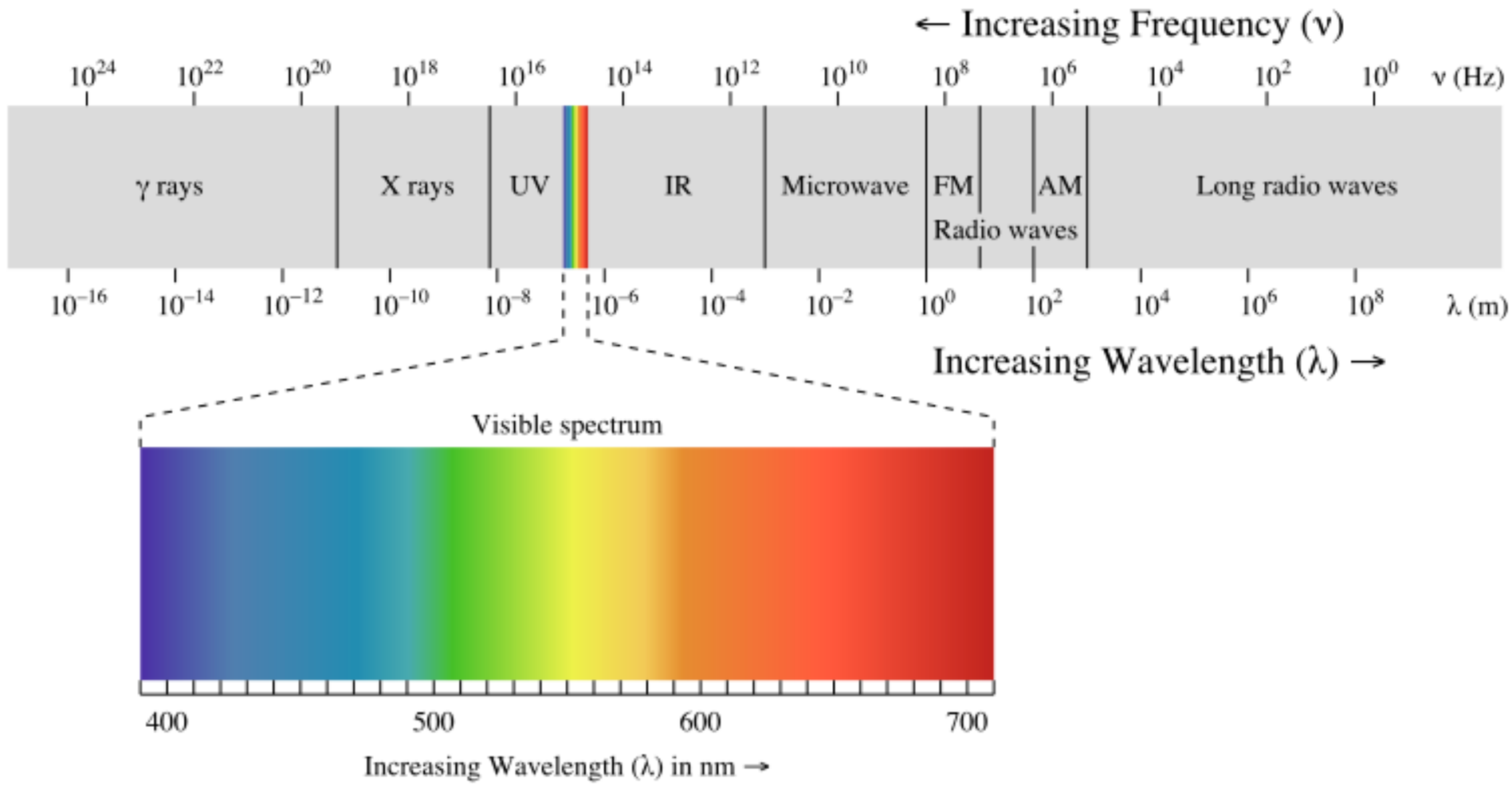


Image credit: Licensed under CC BY-SA 3.0 via Commons
https://commons.wikimedia.org/wiki/File:EM_spectrum.svg#/media/File:EM_spectrum.svg

Lights: How Do They Work?



Cree 11 W LED light bulb (60W incandescent replacement)

Physical process converts energy into photons

- Each photon carries a small amount of energy

Over some amount of time, bulb consumes some amount of energy, Joules

- Some is turned into heat, some into photons

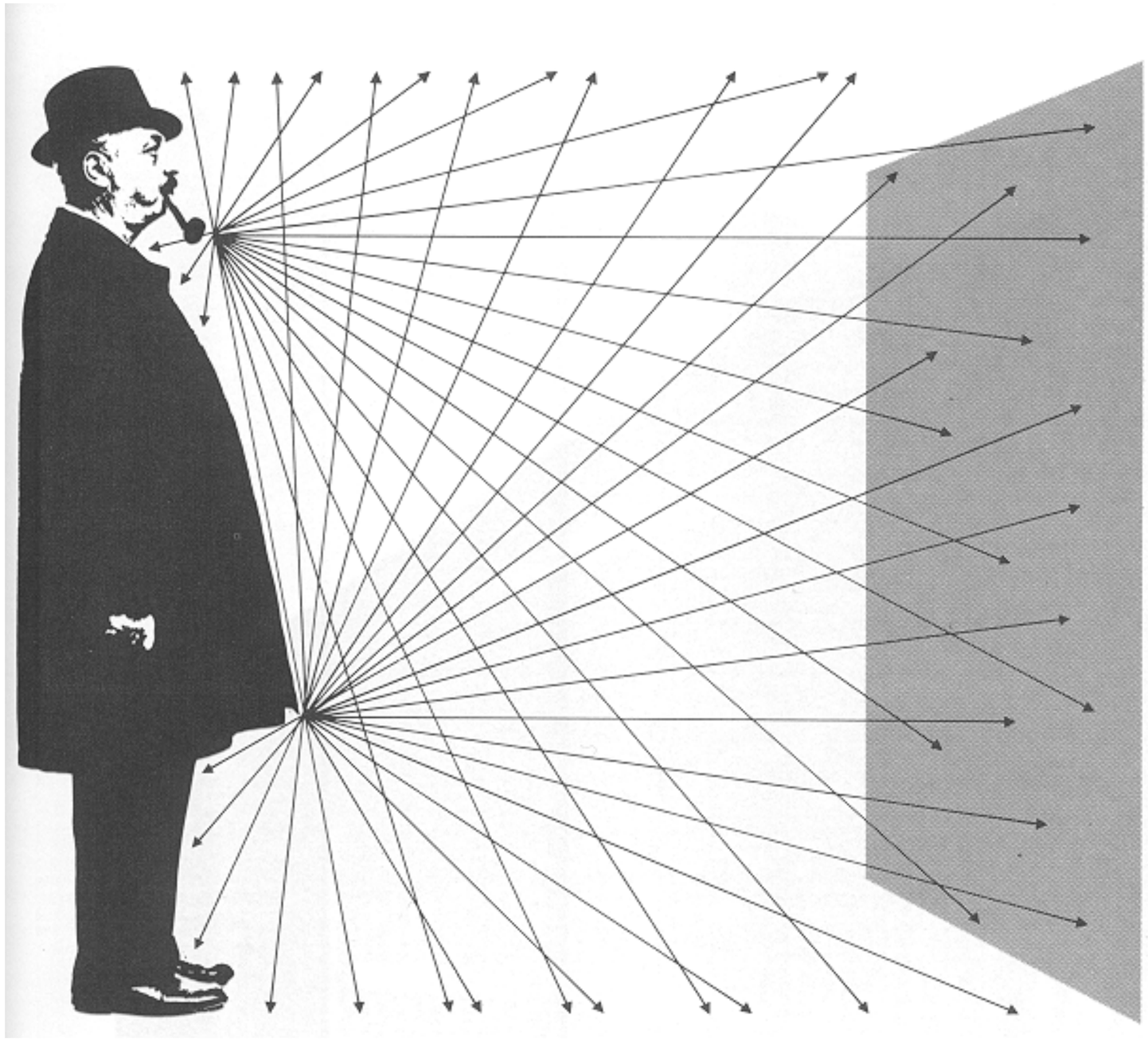
Energy of photons hitting an object ~ exposure

- Film, sensors, sunburn, solar panels, …

Graphics: generally assume “steady state” flow

- Rate of energy consumption is constant, so flux (power) and energy are often interchangeable

Flux – What's the Density of Photons Flowing Through a Sensor?



From London and Upton

Radiant Energy and Flux (Power)

Radiant Energy and Flux (Power)

Definition: Radiant (luminous*) energy is the energy of electromagnetic radiation. It is measured in units of joules, and denoted by the symbol:

$$Q \ [\mathrm{J} = \mathrm{Joule}]$$ **(Watt-Hour)**

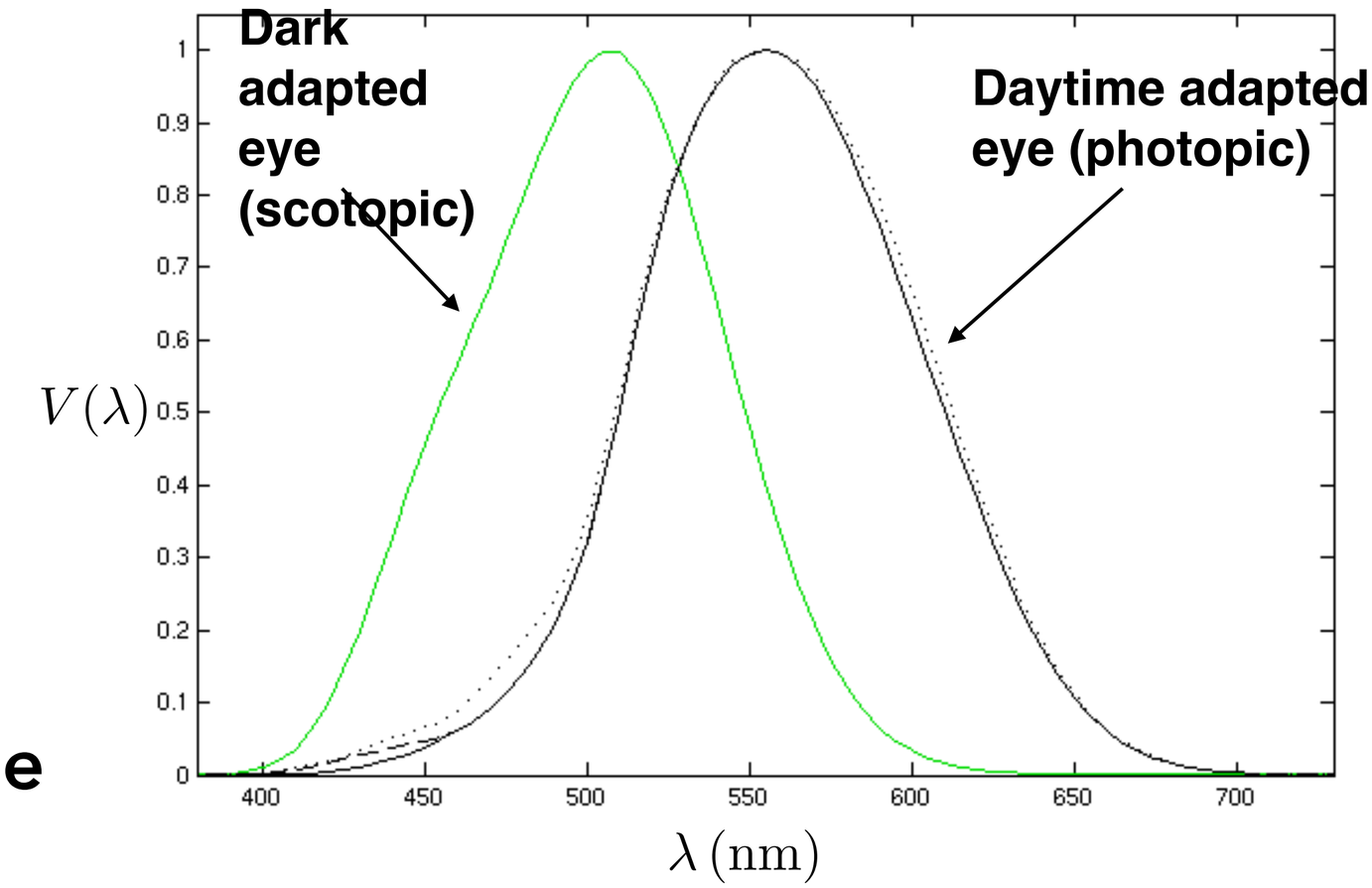
Definition: Radiant (luminous*) flux is the energy emitted, reflected, transmitted or received, per unit time.

$$\Phi \equiv \frac{\mathrm{d}Q}{\mathrm{d}t} \ [\mathrm{W} = \mathrm{Watt}] \ [\mathrm{lm} = \mathrm{lumen}]^{*}$$

*** Definition slides will provide photometric terms in parentheses and give photometric units**

Photometry

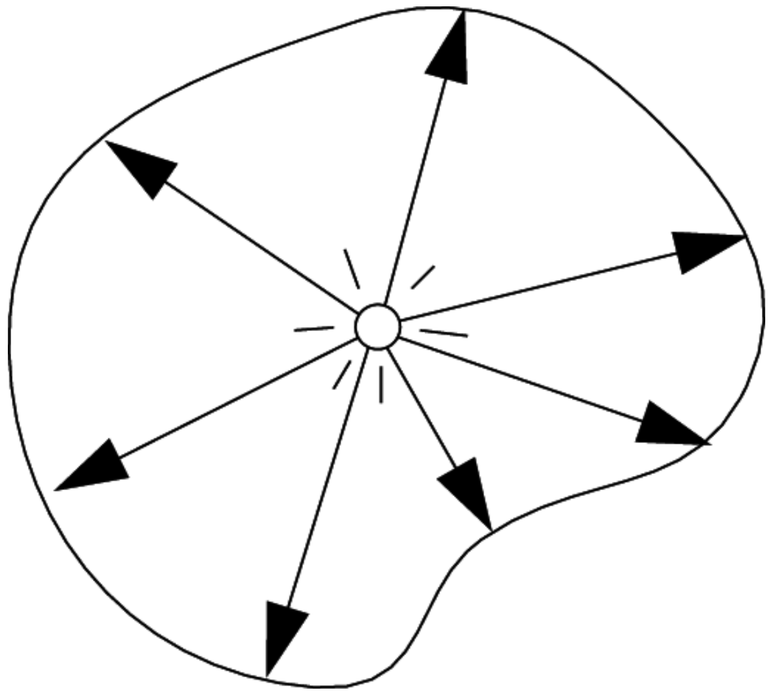
- All radiometric quantities have equivalents in photometry
- Photometry: accounts for response of human visual system
- E.g. Luminous flux Φ_v is the photometric quantity that corresponds to radiant flux Φ_e: integrate radiant flux over all wavelengths, weighted by eye's luminous efficiency curve $V(\lambda)$



https://upload.wikimedia.org/wikipedia/commons/a/a0/Luminosity.png

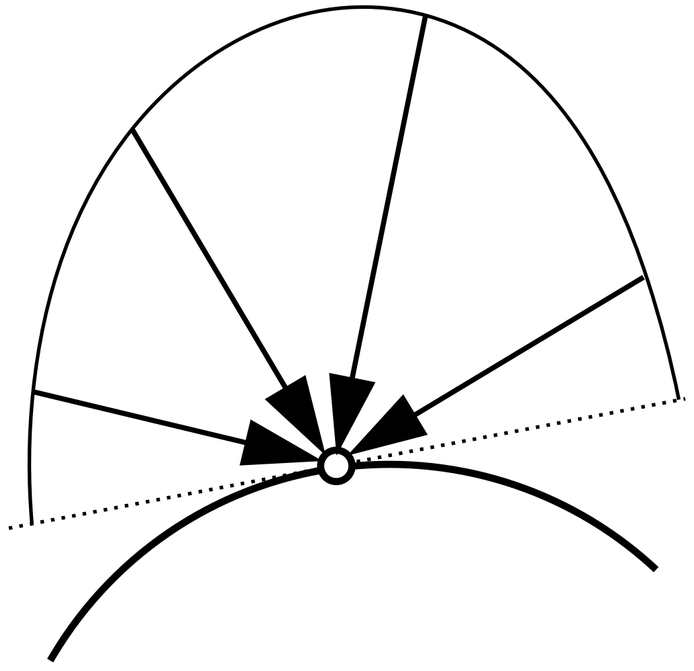
$$\Phi_v = \int_0^\infty \Phi_e(\lambda) V(\lambda)\, d\lambda$$

Example Light Measurements of Interest



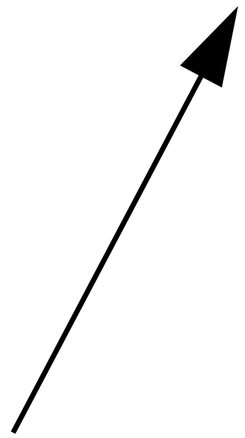
Light Emitted From A Source

"Radiant Intensity"



Light Falling On A Surface

"Irradiance"



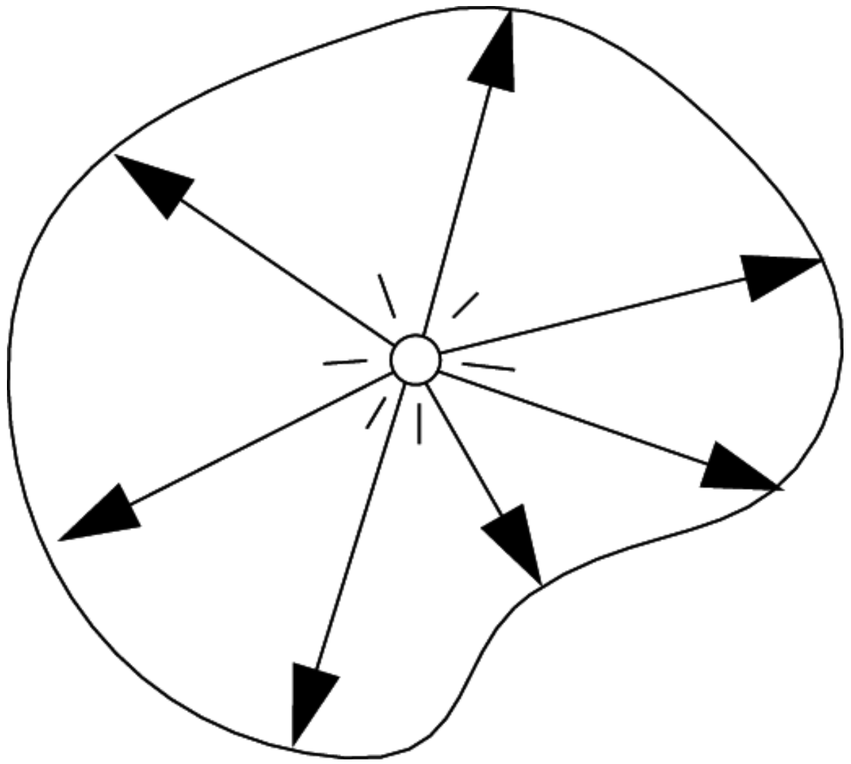
Light Traveling Along A Ray

"Radiance"

Radiant Intensity

Radiant Intensity

Definition: The radiant (luminous) intensity is the power per unit solid angle emitted by a point light source.



$$I(\omega) \equiv \frac{\mathrm{d}\Phi}{\mathrm{d}\omega}$$

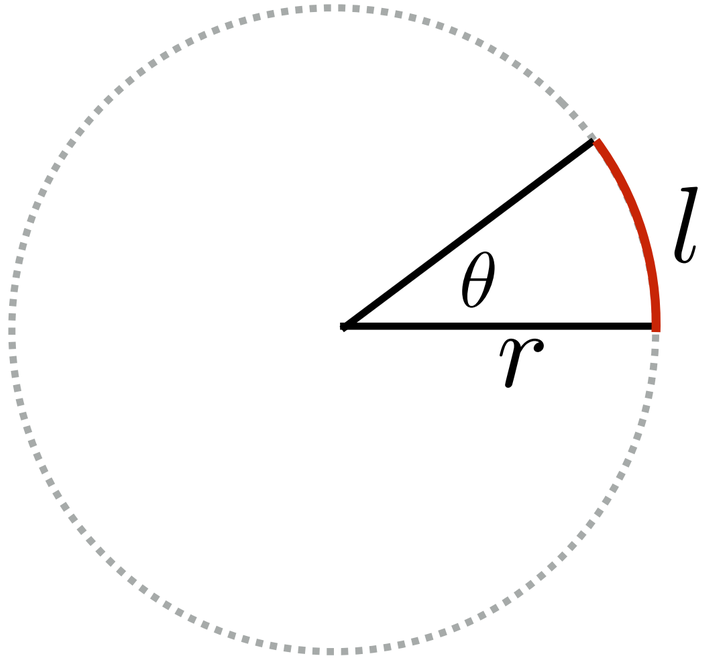
$$\left[\frac{\mathrm{W}}{\mathrm{sr}}\right] \left[\frac{\mathrm{lm}}{\mathrm{sr}} = \mathrm{cd} = \mathrm{candela}\right]$$

The candela is one of the seven SI base units (m, s, mole, A, K, cd, kg)

Angles and Solid Angles

Angle: ratio of subtended arc length on circle to radius

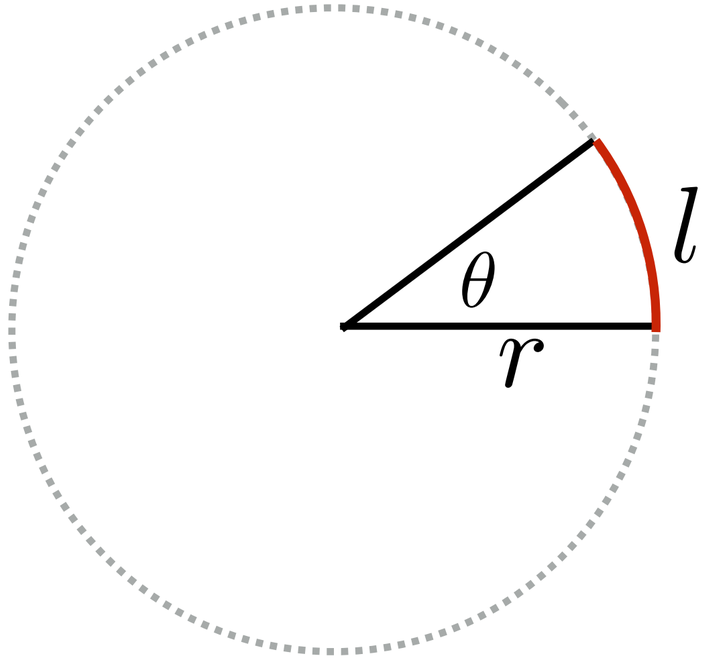
- $\theta = \frac{l}{r}$
- **Circle has** 2π **radians**



Angles and Solid Angles

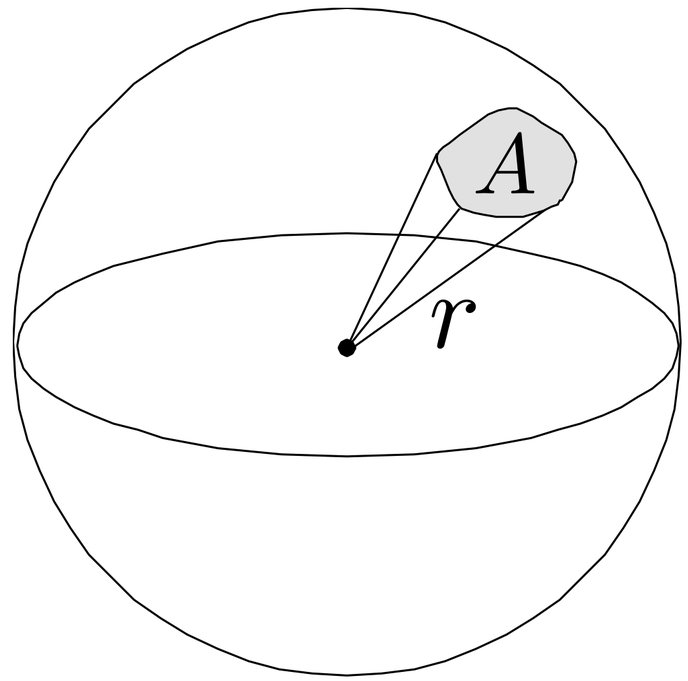
Angle: ratio of subtended arc length on circle to radius

- $\theta = \frac{l}{r}$
- **Circle has** 2π **radians**

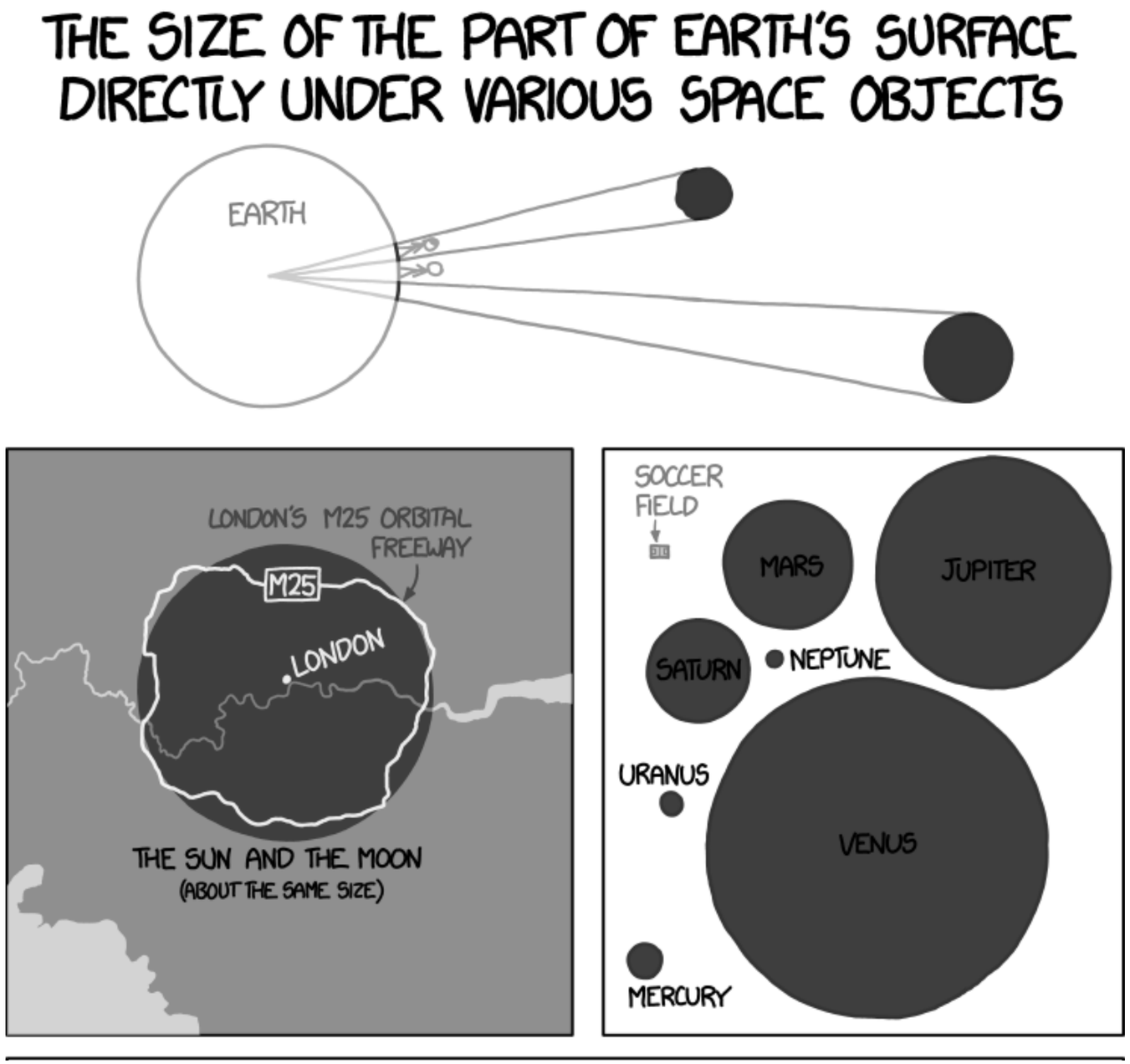


Solid angle: ratio of subtended area on sphere to radius squared

- $\Omega = \frac{A}{r^2}$
- **Sphere has** 4π **steradians**



Solid Angles in Practice

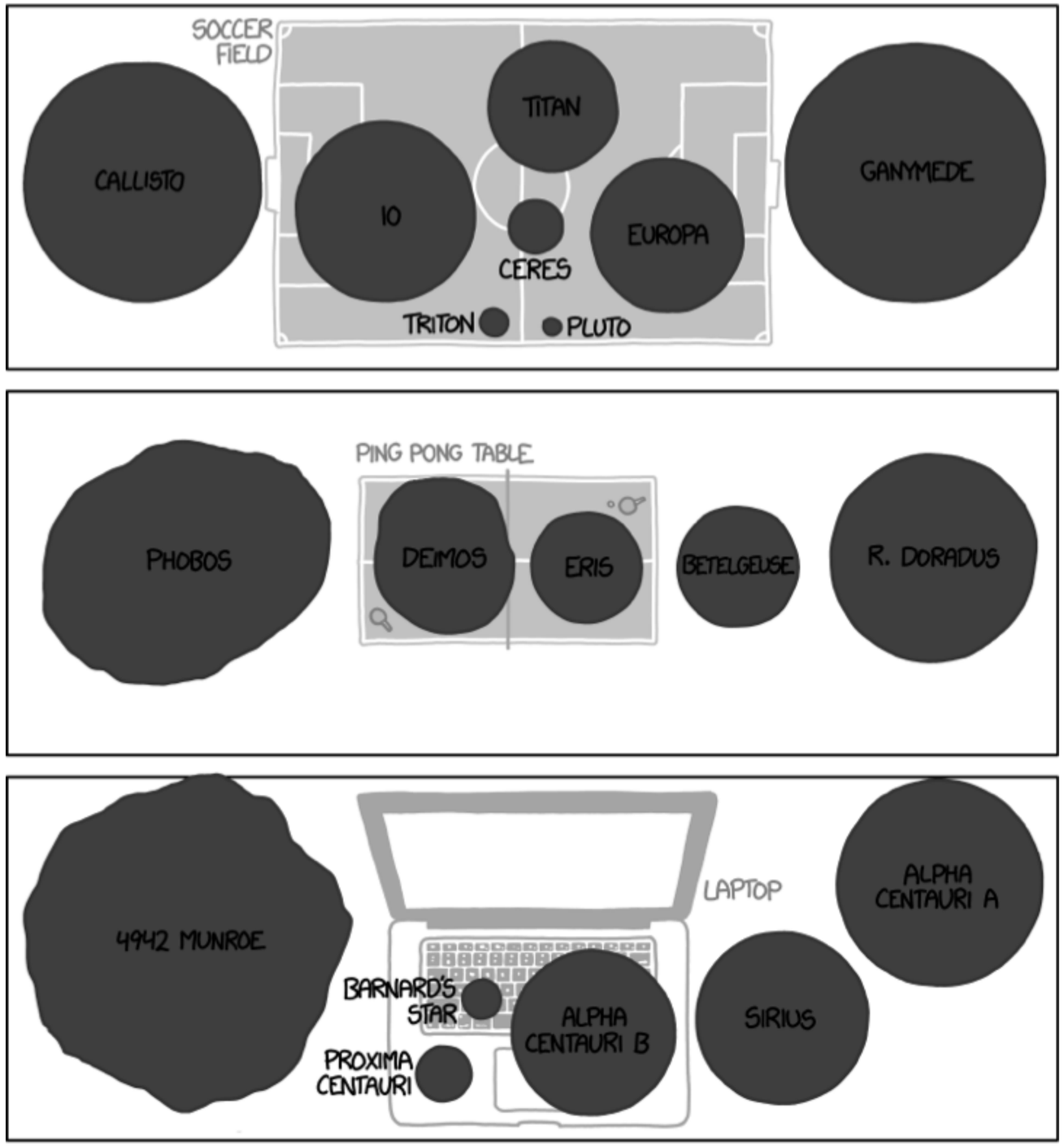


http://xkcd.com/1276/

- Sun and moon both subtend ~60μ sr as seen from earth
- Surface area of earth: ~510M km²
- Projected area:

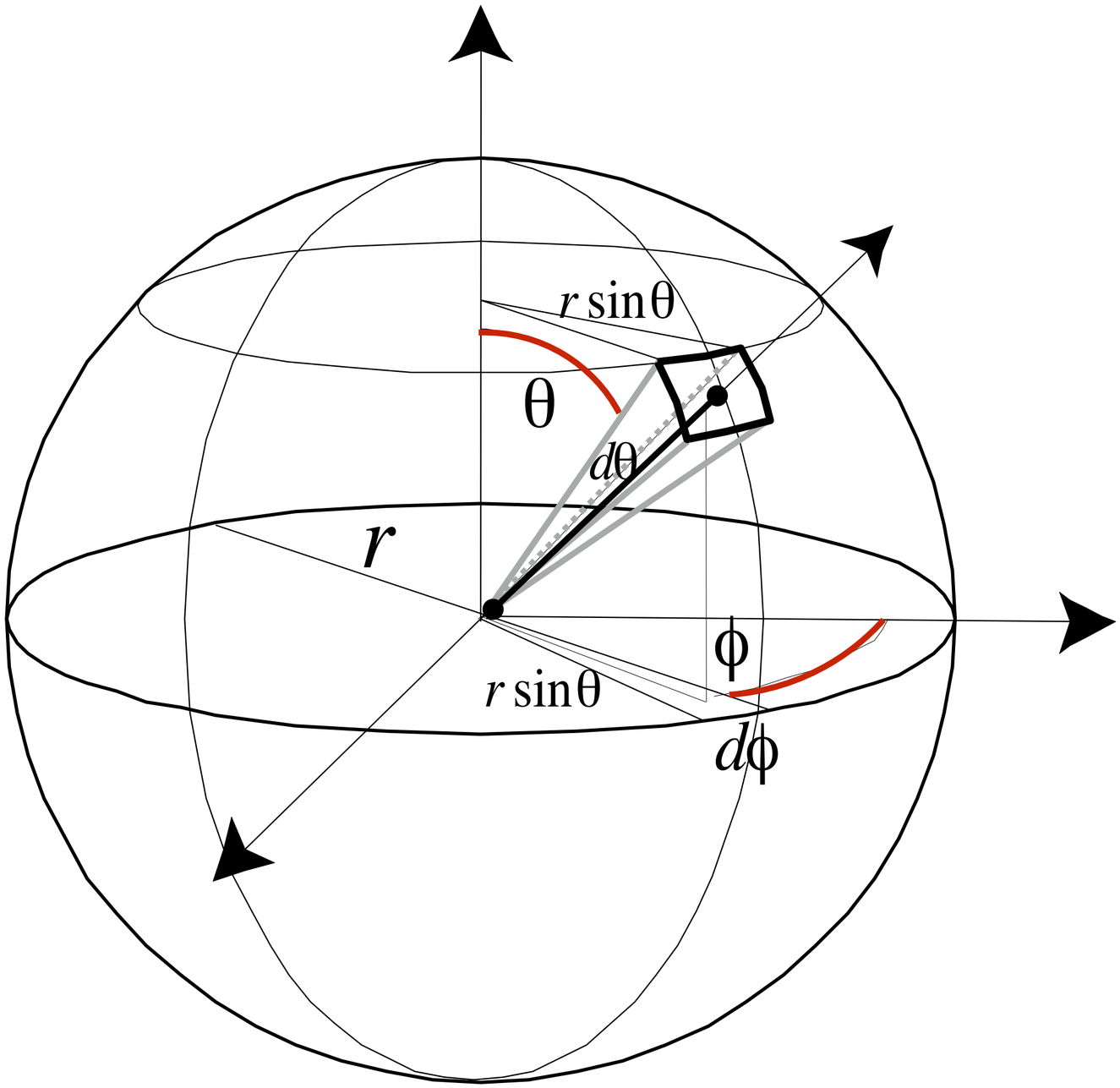
$$510\text{Mkm}^2 \frac{60\mu\text{sr}}{4\pi\text{sr}} = 510\frac{15}{\pi}$$
$$\approx 2400\text{km}^2$$

Solid Angles in Practice

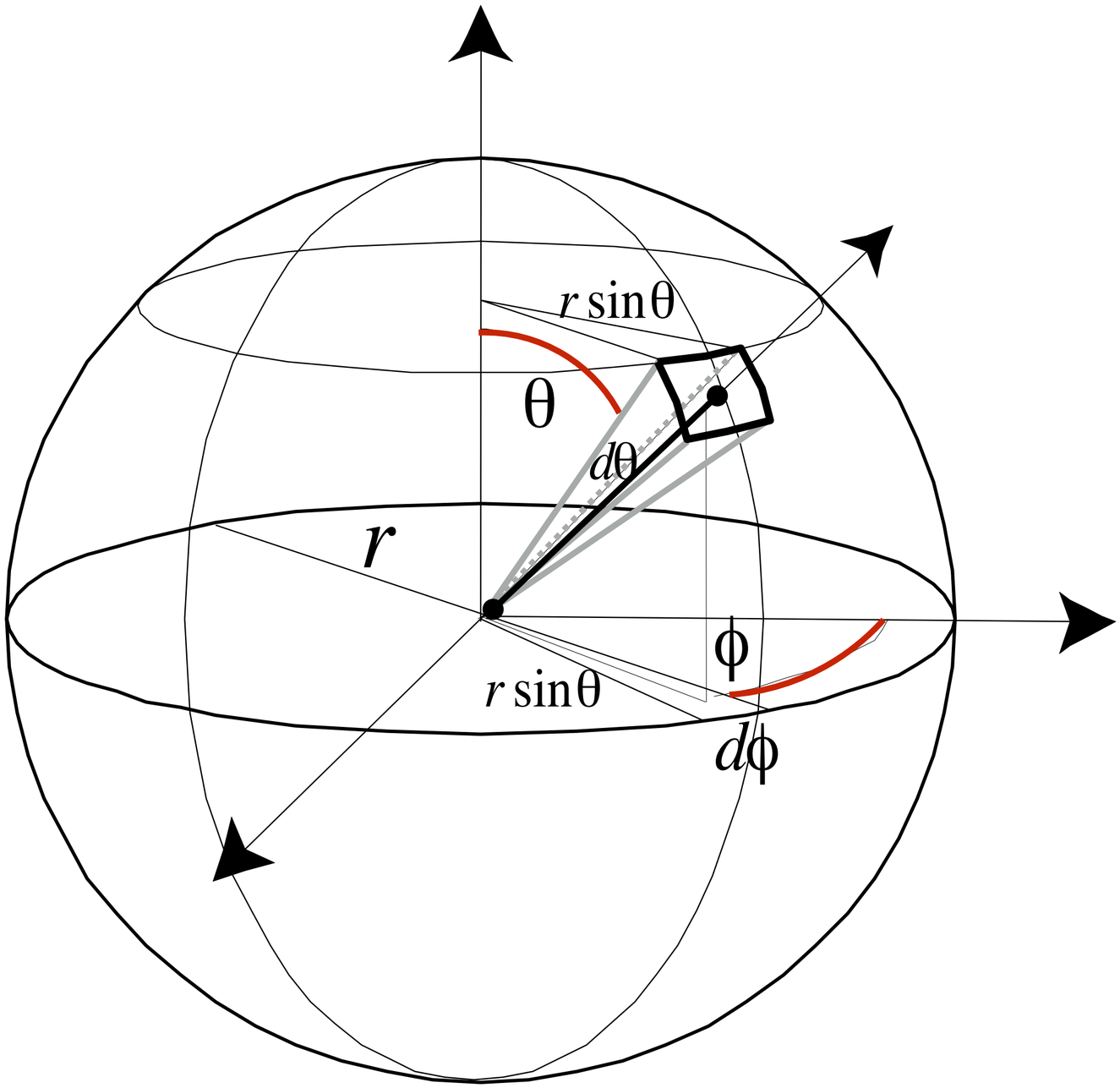


http://xkcd.com/1276/

Differential Solid Angles

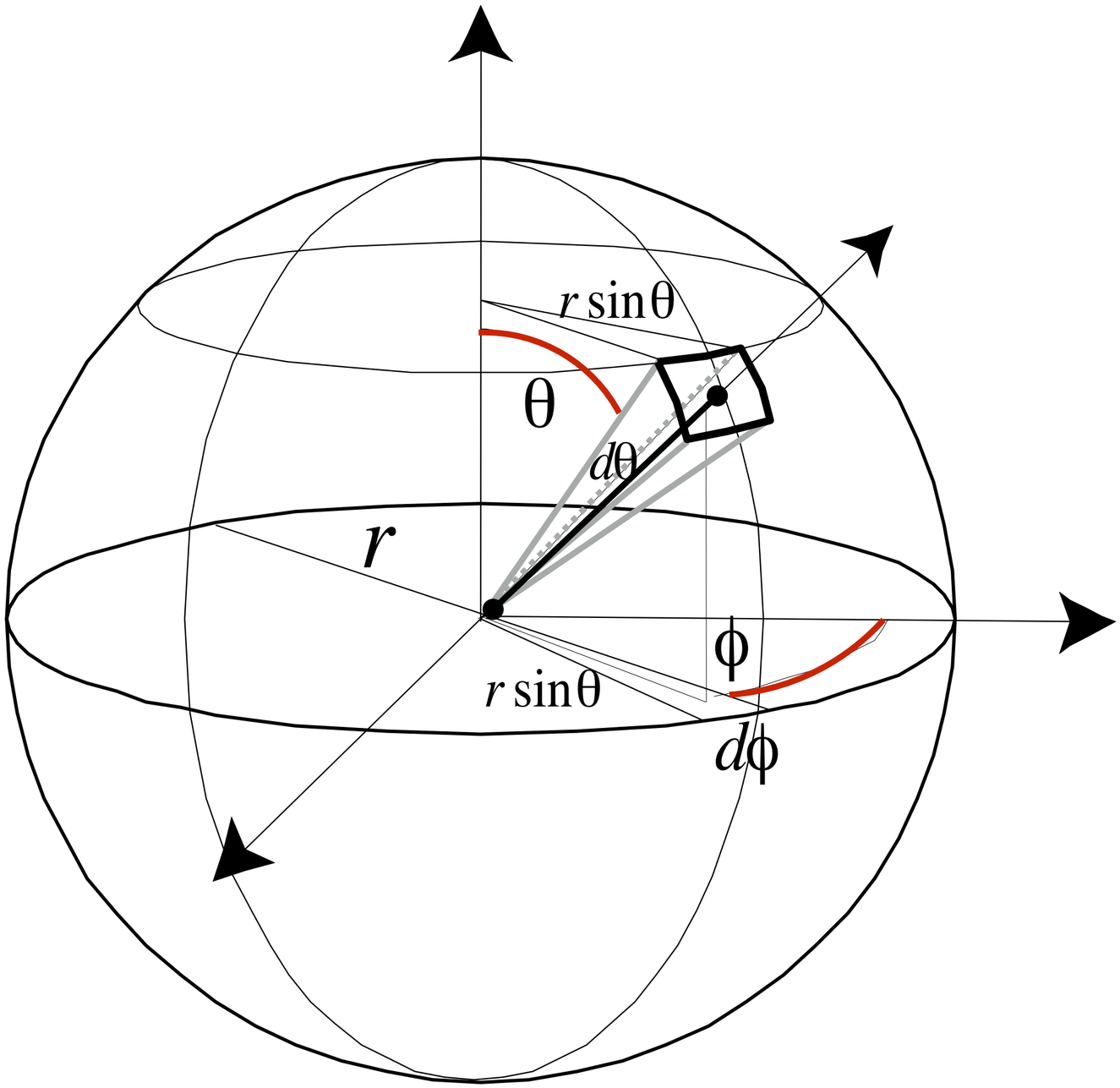


Differential Solid Angles



$$\begin{aligned} \mathrm{d}A &= (r\,\mathrm{d}\theta)(r\sin\theta\,\mathrm{d}\phi) \\ &= r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi \end{aligned}$$

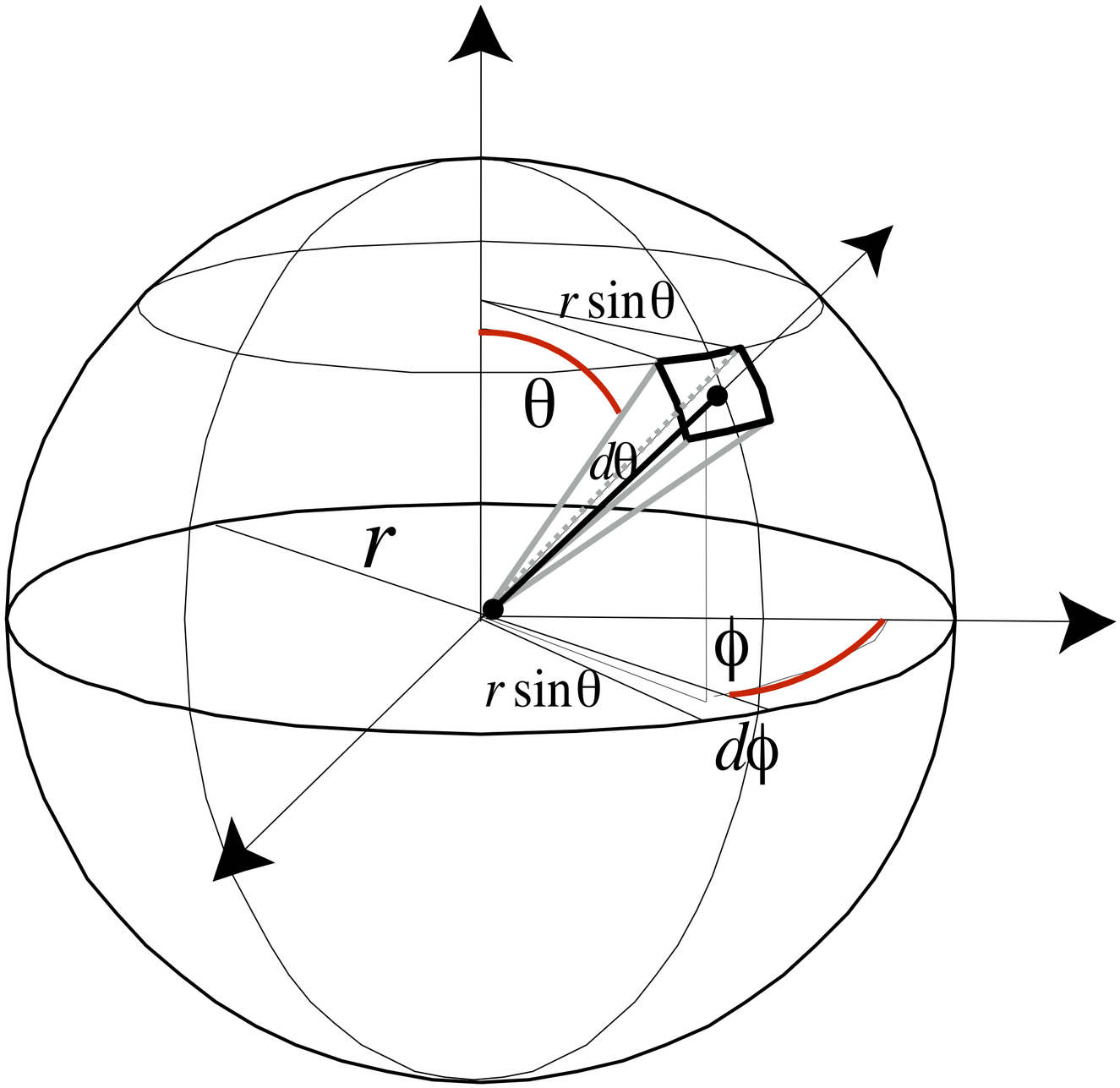
Differential Solid Angles



$$\begin{aligned} \mathrm{d}A =& (r\,\mathrm{d}\theta)(r\sin\theta\,\mathrm{d}\phi) \\ =& r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi \end{aligned}$$

$$\mathrm{d}\omega = \frac{\mathrm{d}A}{r^2} = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$$

Differential Solid Angles

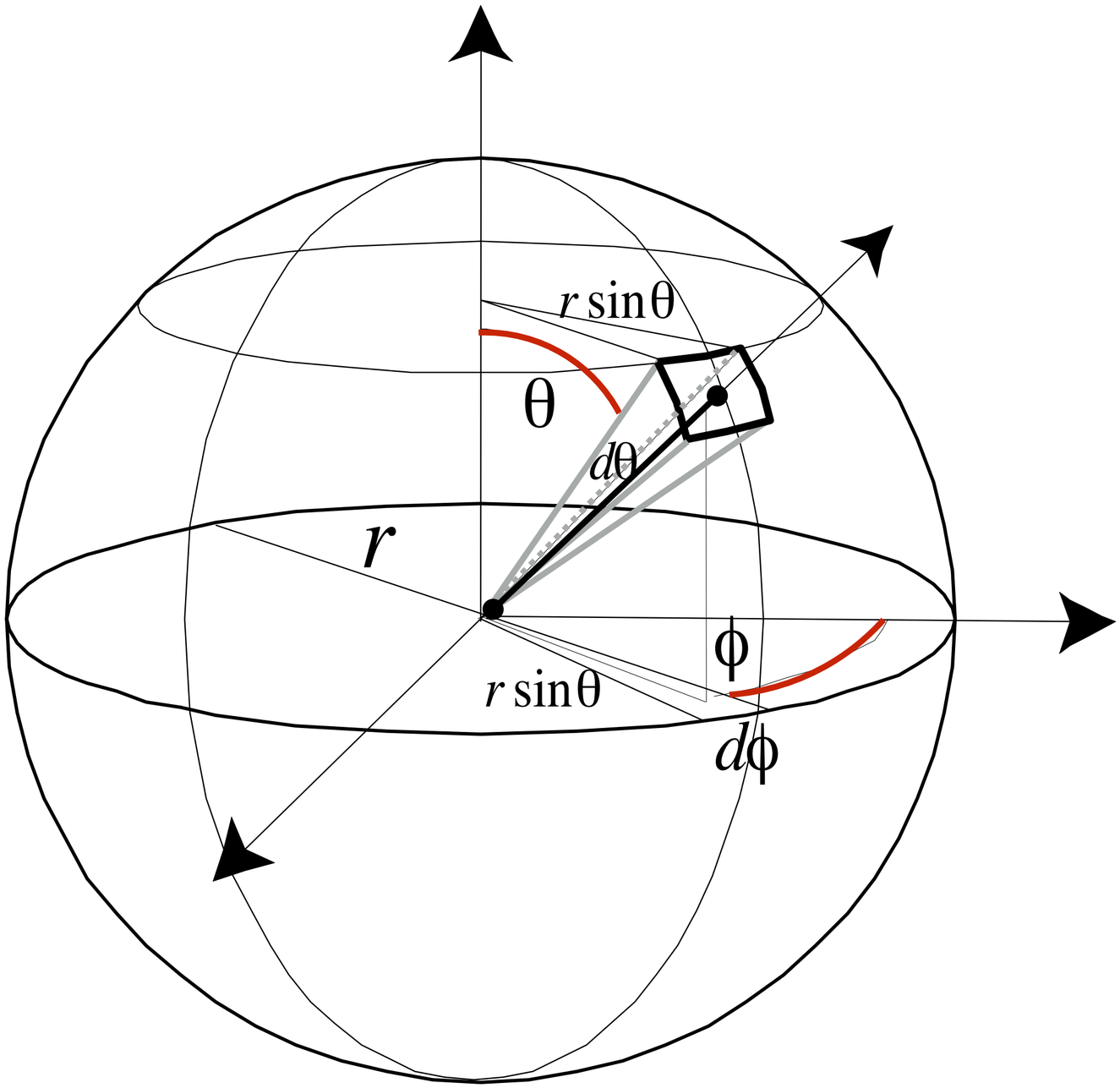


Sphere: S^2

$$\Omega = \int_{S^2} \mathrm{d}\omega$$

$$= \int_0^{2\pi} \int_0^{\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi$$

Differential Solid Angles



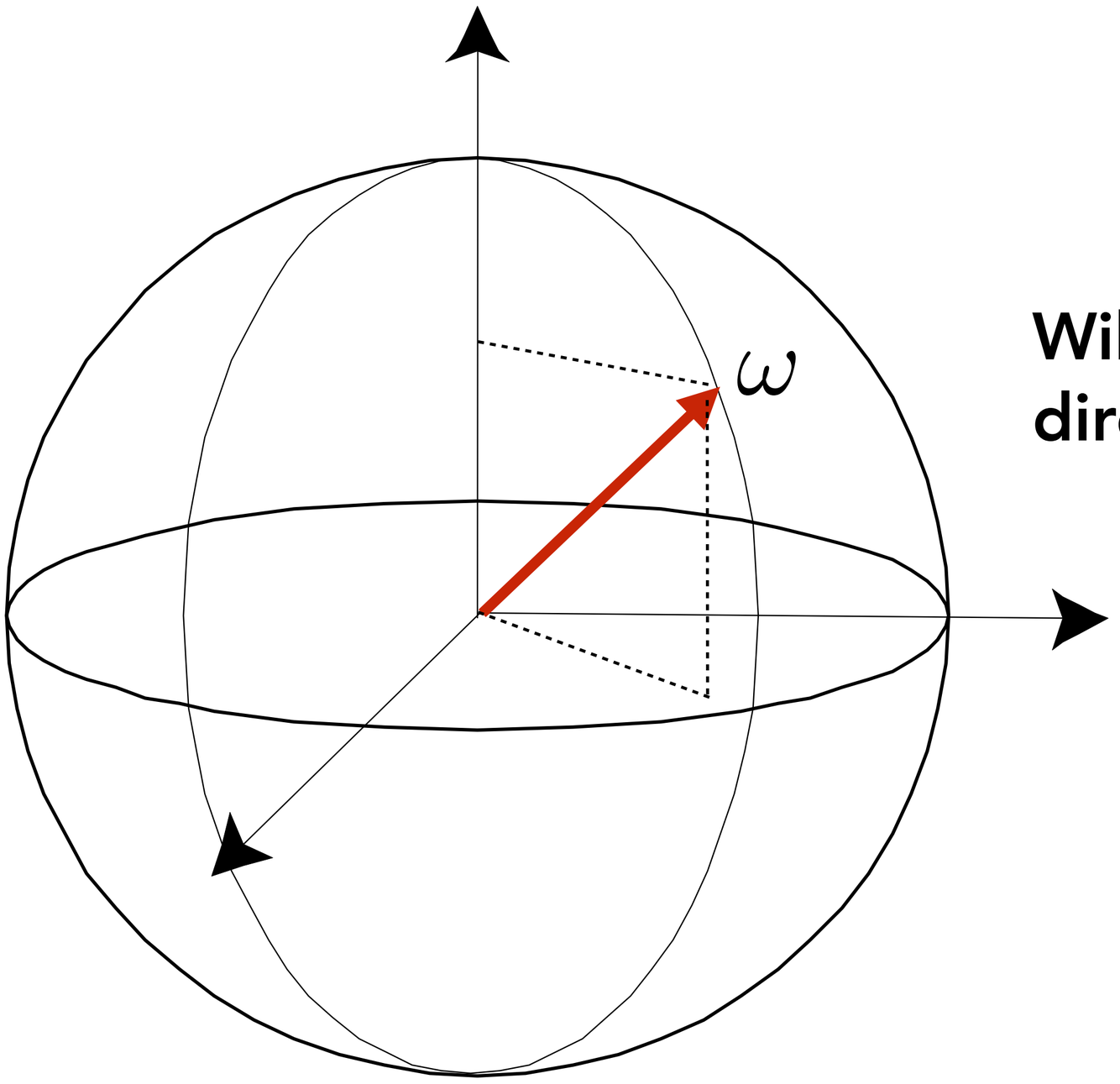
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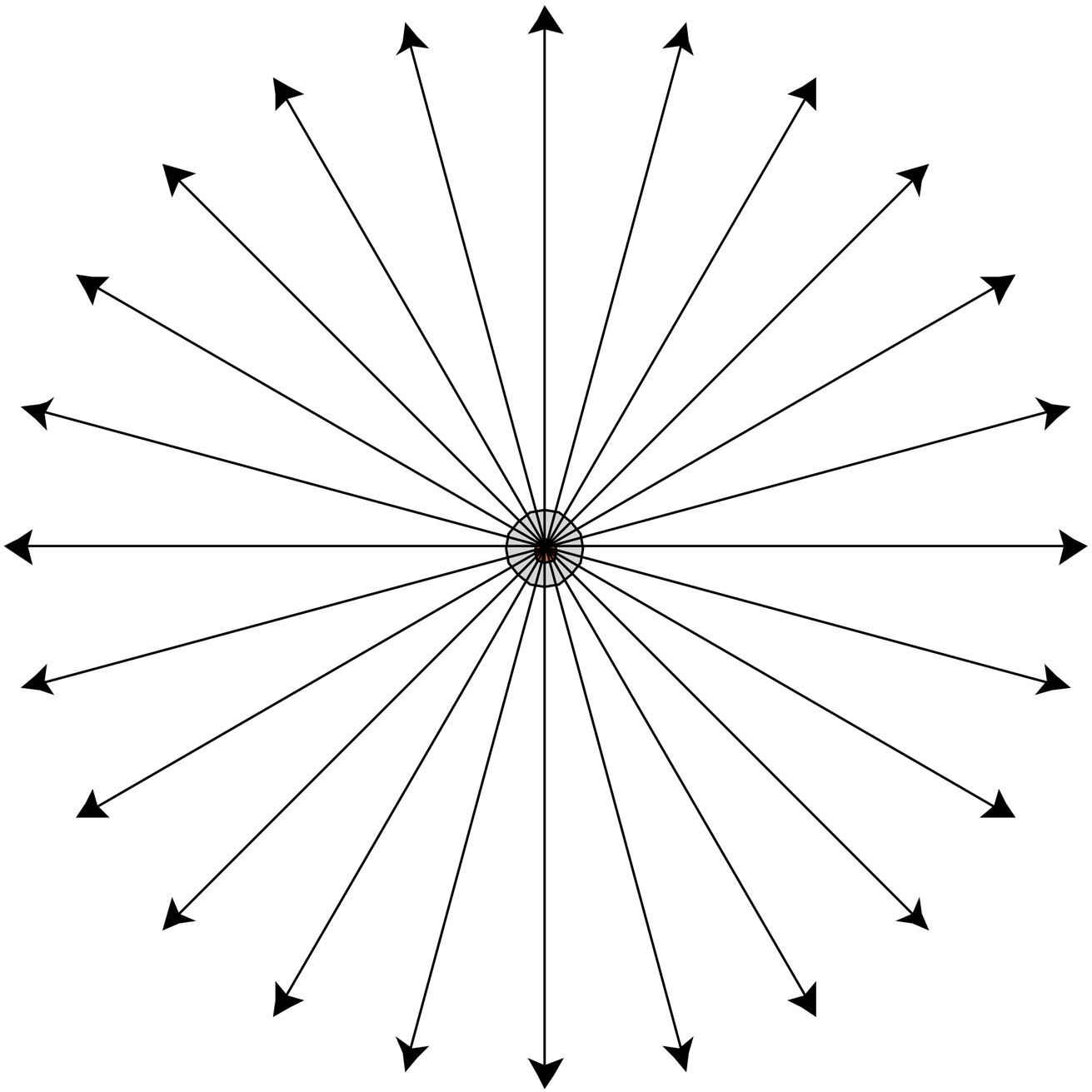
$$= 4\pi$$

ω as a direction vector



Will use ω to denote a direction vector (unit length)

Isotropic Point Source



$$\Phi = \int_{S^2} I \, \mathrm{d}\omega$$

$$= 4\pi I$$

$$I = \frac{\Phi}{4\pi}$$

Modern LED Light

Output: 815 lumens

(11W LED replacement for 60W incandescent)

Luminous intensity?

Assume isotropic:
Intensity = 815 lumens / 4pi sr
= 65 candelas



Spectral Power Distribution - More in Color Lectures

Describes distribution of energy by wavelength

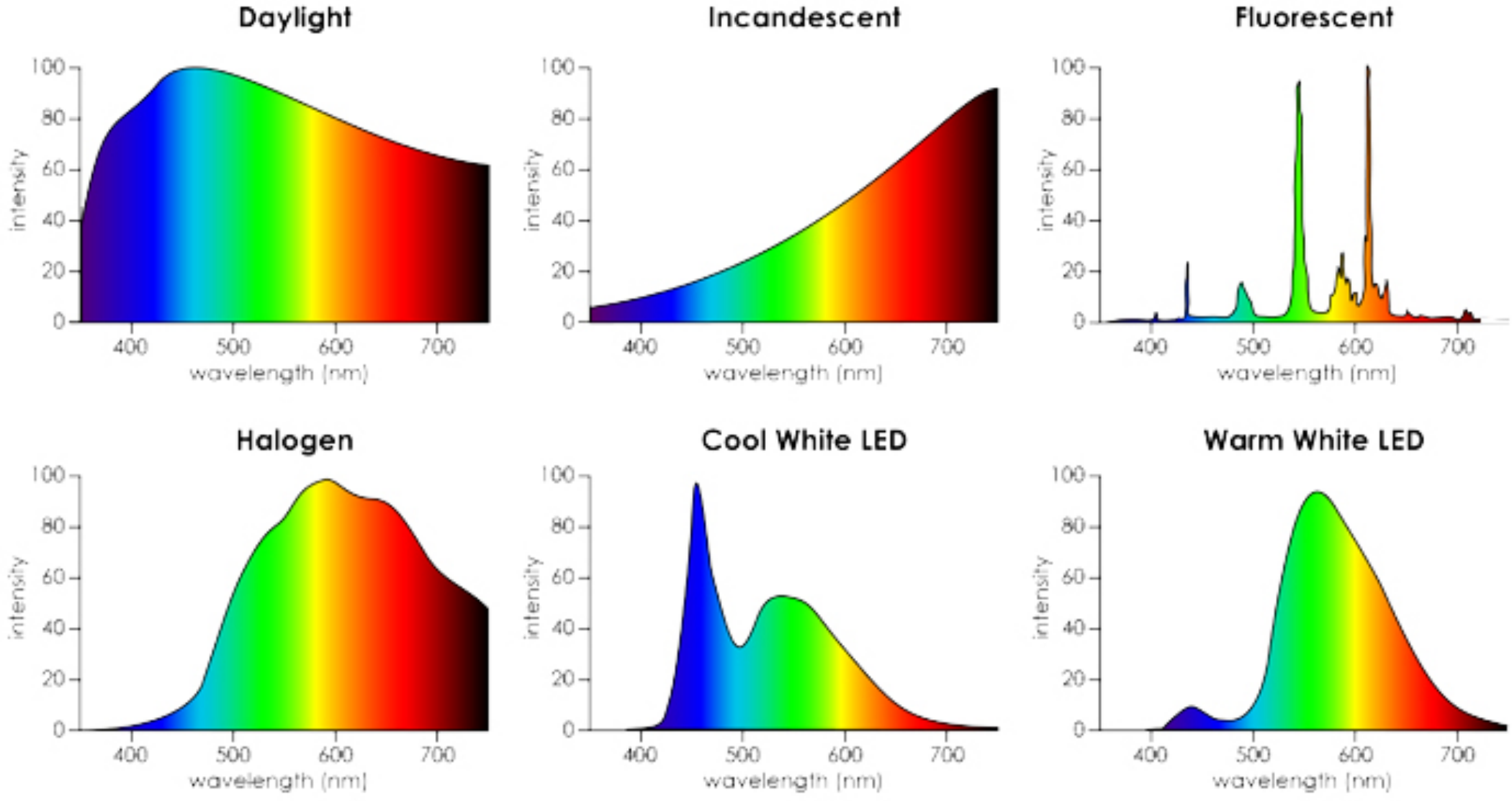
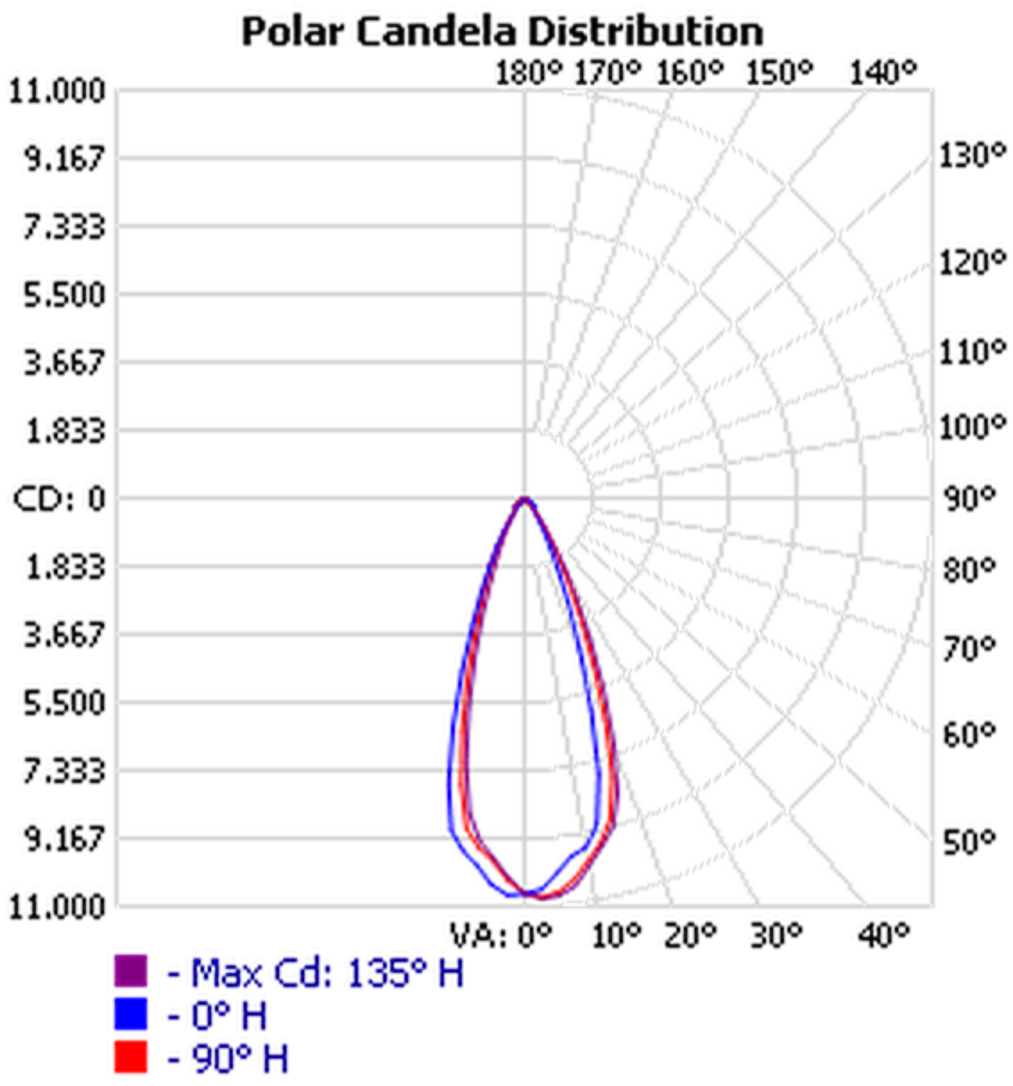
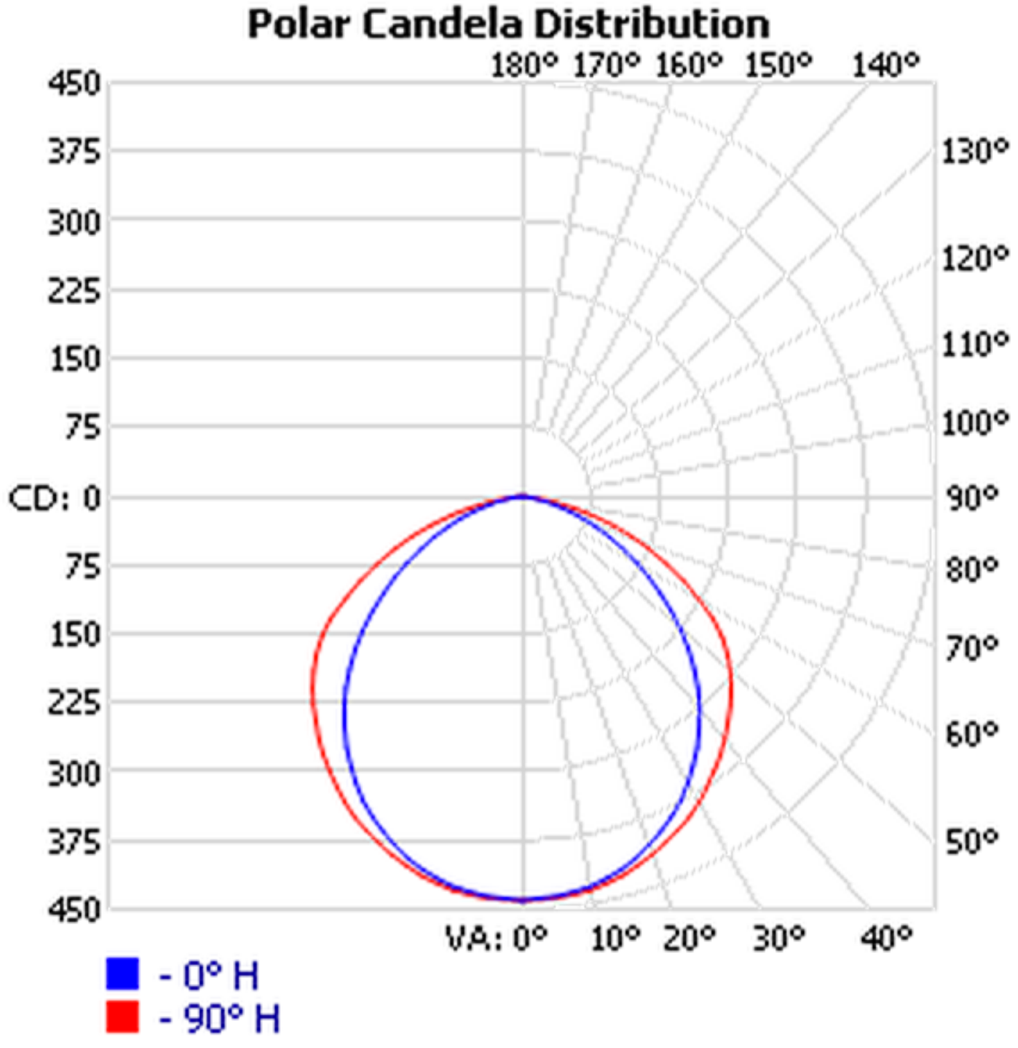


Figure credit: admesy ADVANCED MEASUREMENT SYSTEMS

Goniometric Diagrams



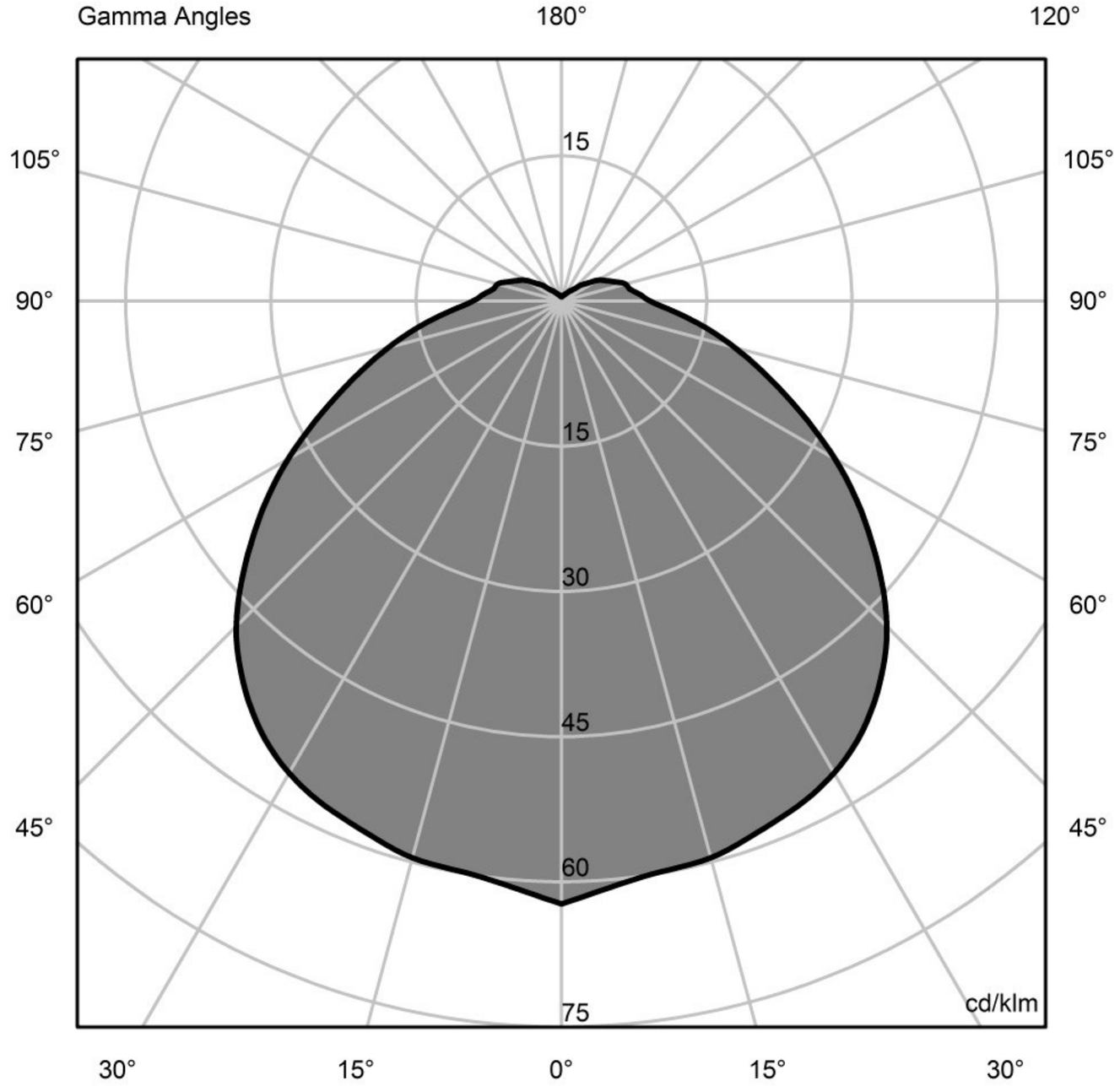


http://www.visual-3d.com/tools/photometricviewer/

Light Fixture Measurements - Goniometric Diagram



Poul Henningsen's Artichoke Lamp



Polar AKA Goniometric Diagram

http://www.louispoulsen.com/

CS184/284A

F26 - O'Brien



http://www.louispoulsen.com/

PH Artichoke Lamps in Rivercenter for the Performing Arts, Georgia

Rendering with Goniometric Diagrams

http://resources.mpi-inf.mpg.de/atrium/

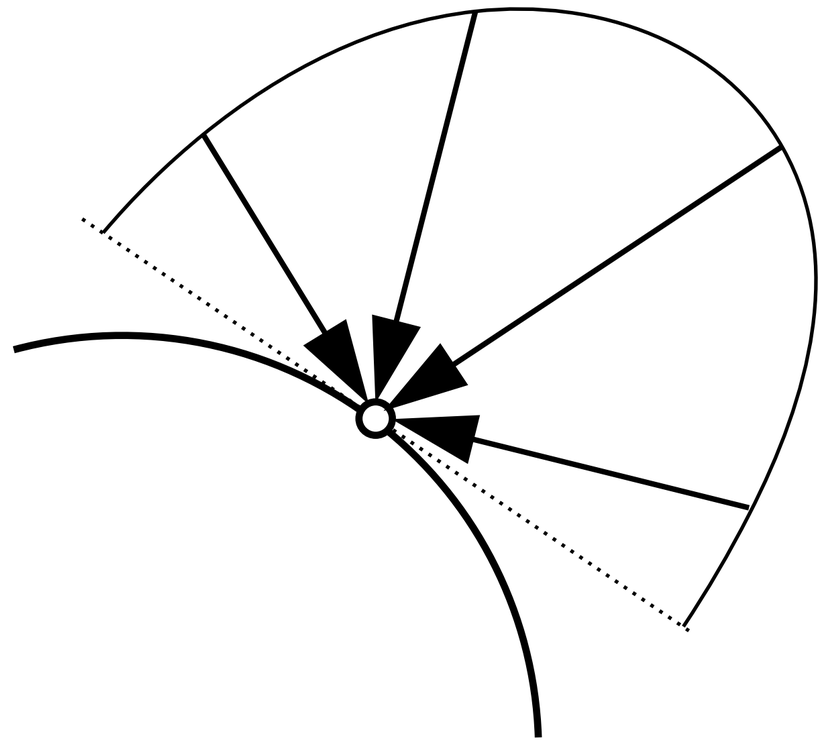
Irradiance

Irradiance

Definition: The irradiance (illuminance) is the power per unit area incident on a surface point.

$$E(\mathbf{x}) \equiv \frac{\mathrm{d}\Phi(\mathbf{x})}{\mathrm{d}A}$$

$$\left[\frac{\mathrm{W}}{\mathrm{m}^2}\right] \quad \left[\frac{\mathrm{lm}}{\mathrm{m}^2} = \mathrm{lux}\right]$$



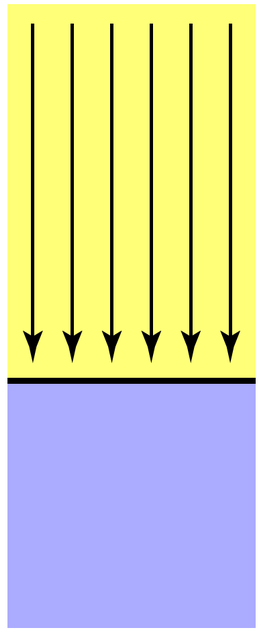
Typical Values of Illuminance [lm/m²]

Brightest sunlight	120,000	lux
Overcast day (midday)	15,000	
Interior near window (daylight)	1,000	
Residential artificial lighting	300	
Sunrise / sunset	40	
Illuminated city street	10	
Moonlight (full)	0.02	
Starlight	0.0003	



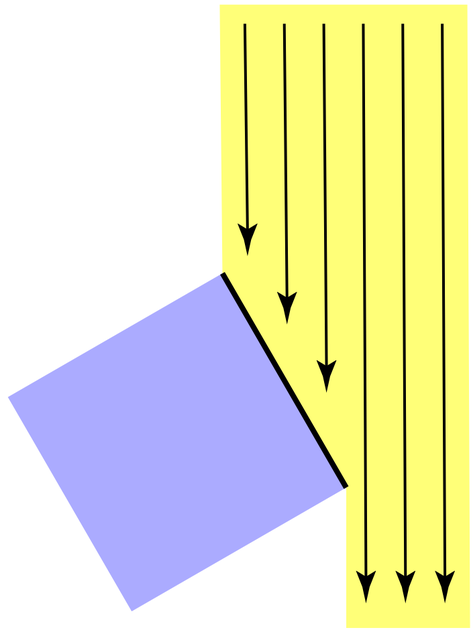
Light meter

Lambert's Cosine Law



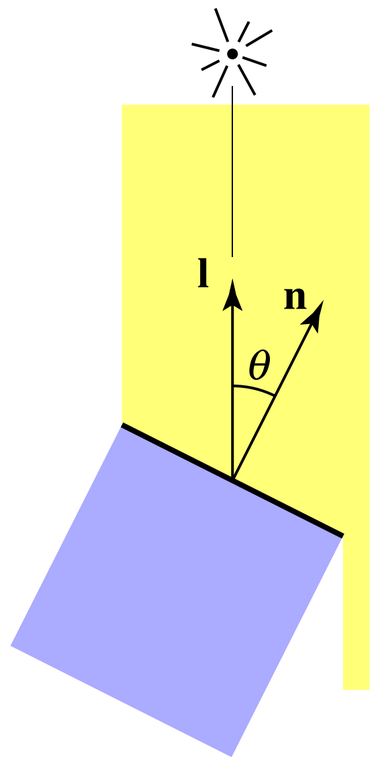
Top face of cube receives a certain amount of power

$$E = \frac{\Phi}{A}$$



Top face of 60° rotated cube receives half power

$$E = \frac{1}{2}\frac{\Phi}{A}$$

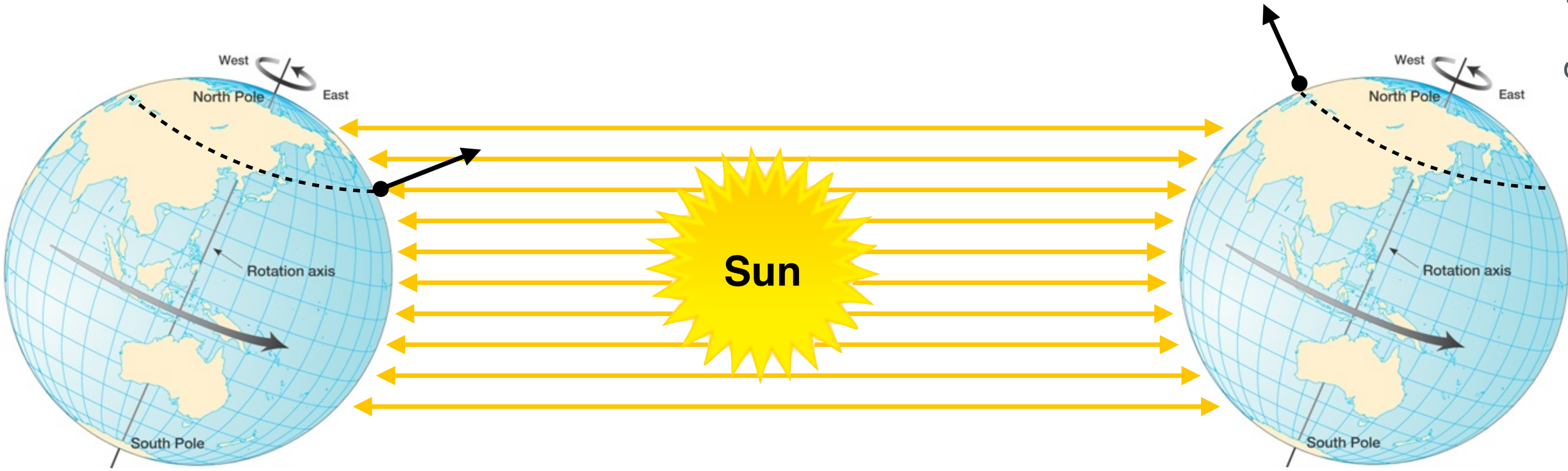


In general, power per unit area is proportional to $\cos\theta = l \cdot n$

$$E = \frac{\Phi}{A}\cos\theta$$

Irradiance at surface is proportional to cosine of angle between light direction and surface normal.

Why Do We Have Seasons?



[Image credit: Pearson Prentice Hall]

Summer
(Northern hemisphere)

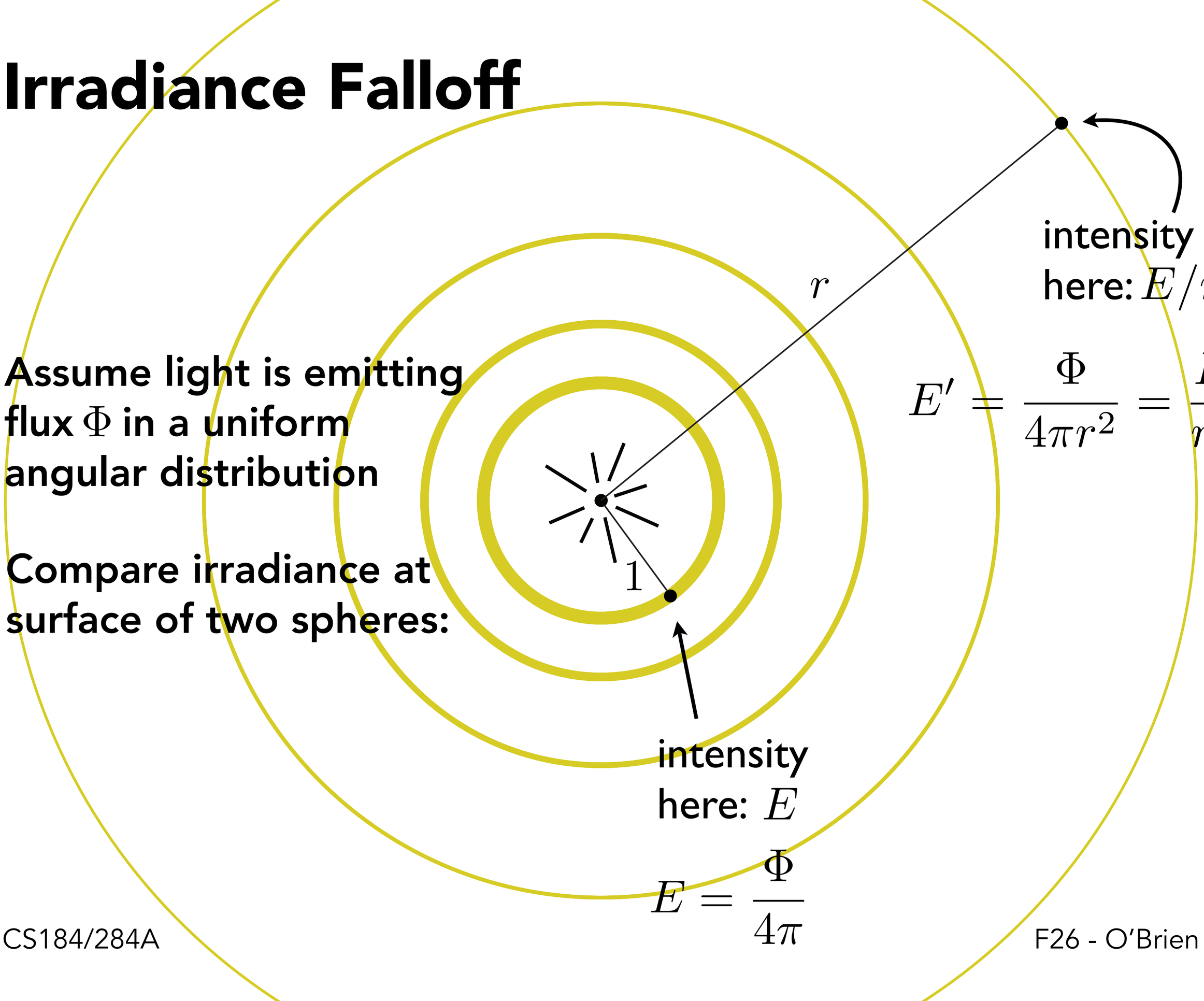
Winter
(Northern hemisphere)

Earth's axis of rotation: ~23.5° off axis

Irradiance Falloff

Assume light is emitting flux Φ in a uniform angular distribution

Compare irradiance at surface of two spheres:



intensity here: E/r^2

$$E' = \frac{\Phi}{4\pi r^2} = \frac{E}{r^2}$$

intensity here: E

$$E = \frac{\Phi}{4\pi}$$

Radiance

Radiance

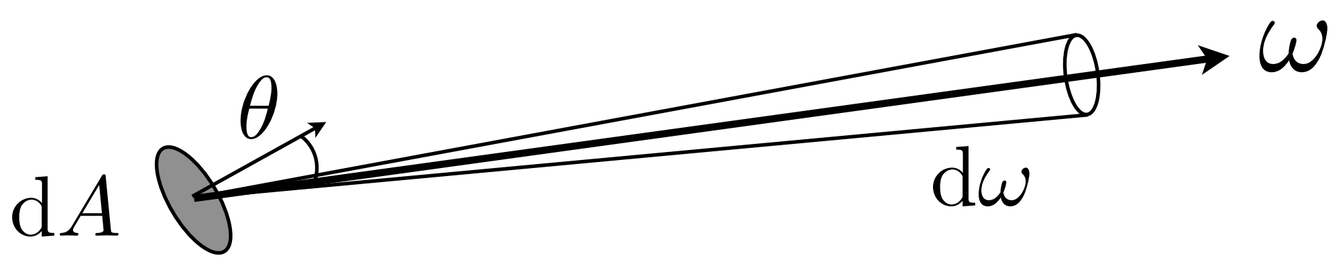


Light Traveling Along A Ray

1. Radiance is the fundamental field quantity that describes the distribution of light in an environment
   - Radiance is the quantity associated with a ray
   - Rendering is all about computing radiance
2. Radiance is invariant along a ray in a vacuum

Surface Radiance

Definition: The radiance (luminance) is the *power* emitted, reflected, transmitted or received by a surface, per unit solid angle, per unit projected area.



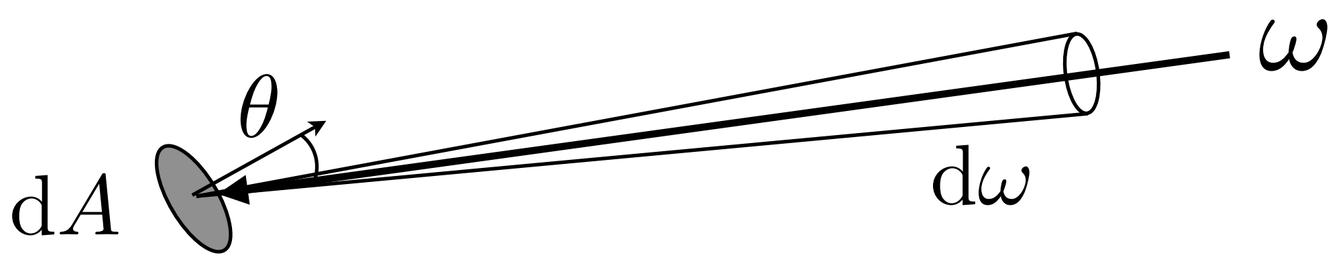
$$L(\mathrm{p},\omega) \equiv \frac{\mathrm{d}^2\Phi(\mathrm{p},\omega)}{\mathrm{d}\omega\,\mathrm{d}A\cos\theta}$$

$\cos\theta$ accounts for projected surface area

$$\left[\frac{\mathrm{W}}{\mathrm{sr\,m^2}}\right] \left[\frac{\mathrm{cd}}{\mathrm{m^2}} = \frac{\mathrm{lm}}{\mathrm{sr\,m^2}} = \mathrm{nit}\right]$$

Incident Surface Radiance

Equivalent: Incident surface radiance (luminance) is the *irradiance* per unit solid angle arriving at the surface.

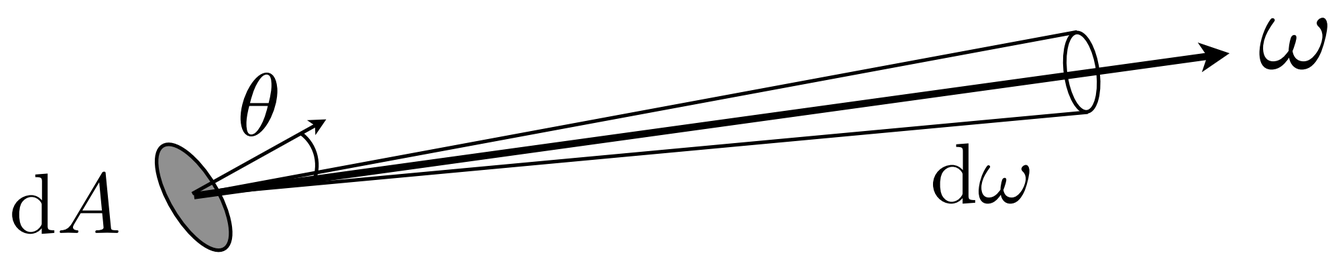


$$L(\mathrm{p}, \omega) = \frac{\mathrm{d}E(\mathrm{p})}{\mathrm{d}\omega \cos\theta}$$

i.e. it is the light arriving at the surface along a given ray (point on surface and incident direction).

Exiting Surface Radiance

Equivalent: Exiting surface radiance (luminance) is the *radiant intensity* per unit projected area leaving the surface.

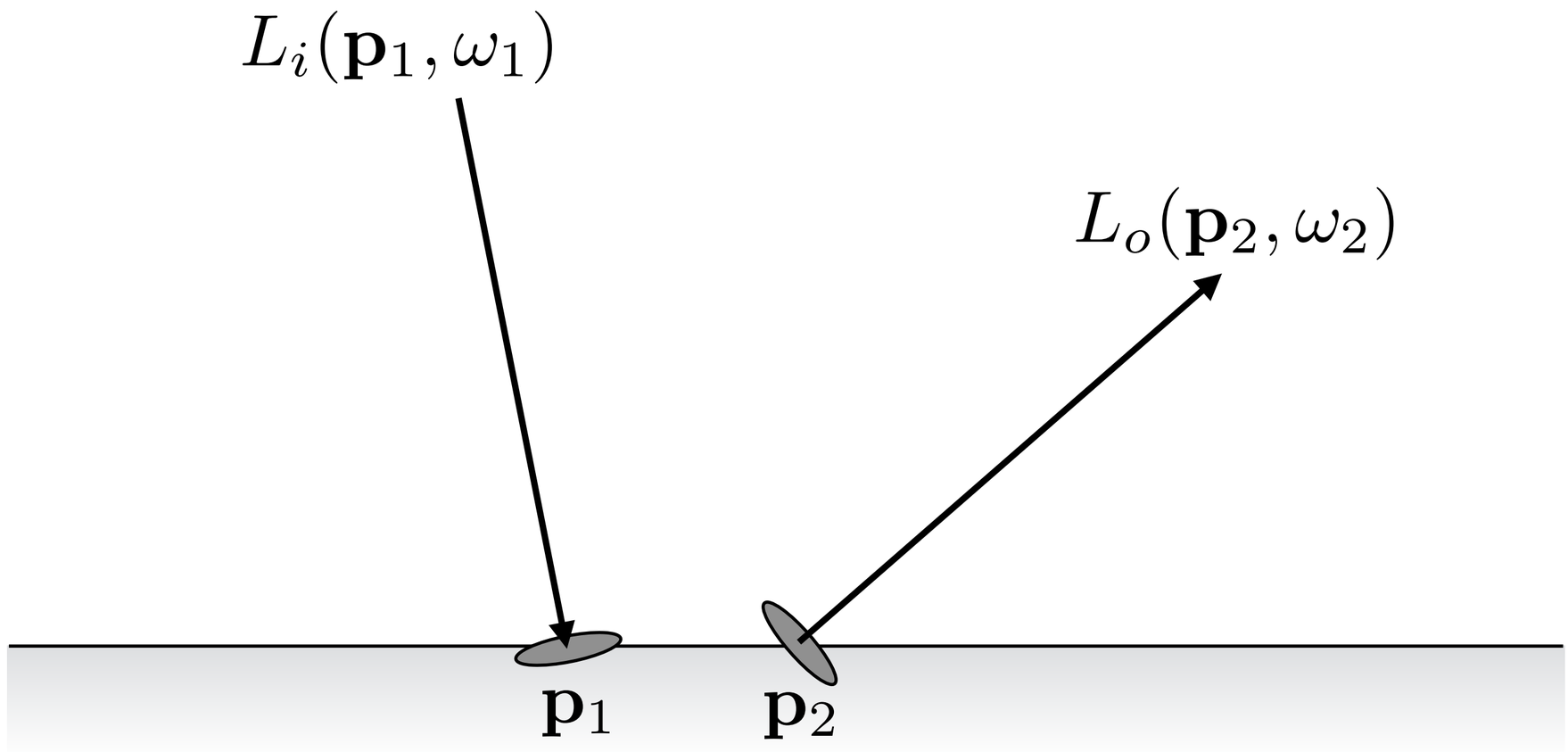


$$L(\mathrm{p}, \omega) = \frac{\mathrm{d}I(\mathrm{p}, \omega)}{\mathrm{d}A \cos\theta}$$

e.g. for an area light it is the light emitted along a given ray (point on surface and exit direction).

Incident & Exiting Surface Radiance Differ!

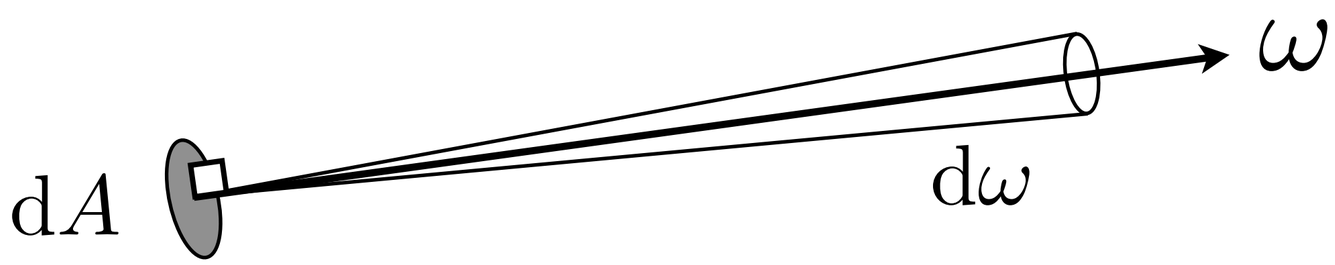
Need to distinguish between incident radiance and exitant radiance functions at a point on a surface



In general: $L_i(\mathbf{p}, \omega) \neq L_o(\mathbf{p}, \omega)$

Field Radiance or Light Field

Definition: The field radiance (luminance) at a point in space in a given direction is the power per unit solid angle per unit area perpendicular to the direction.



Typical Values of Luminance [cd/m²]

Surface of the sun	2,000,000,000 nits
Sunlight clouds	30,000
Clear sky	3,000
Cellphone display	500
Overcast sky	300
Scene at sunrise	30
Scene lit by moon	0.001
Threshold of vision	0.000001

Calculating with Radiance

Irradiance from the Environment

Computing flux per unit area on surface, due to incoming light from all directions.

$$dE(\mathrm{p},\omega) = L_i(\mathrm{p},\omega)\cos\theta\,\mathrm{d}\omega$$

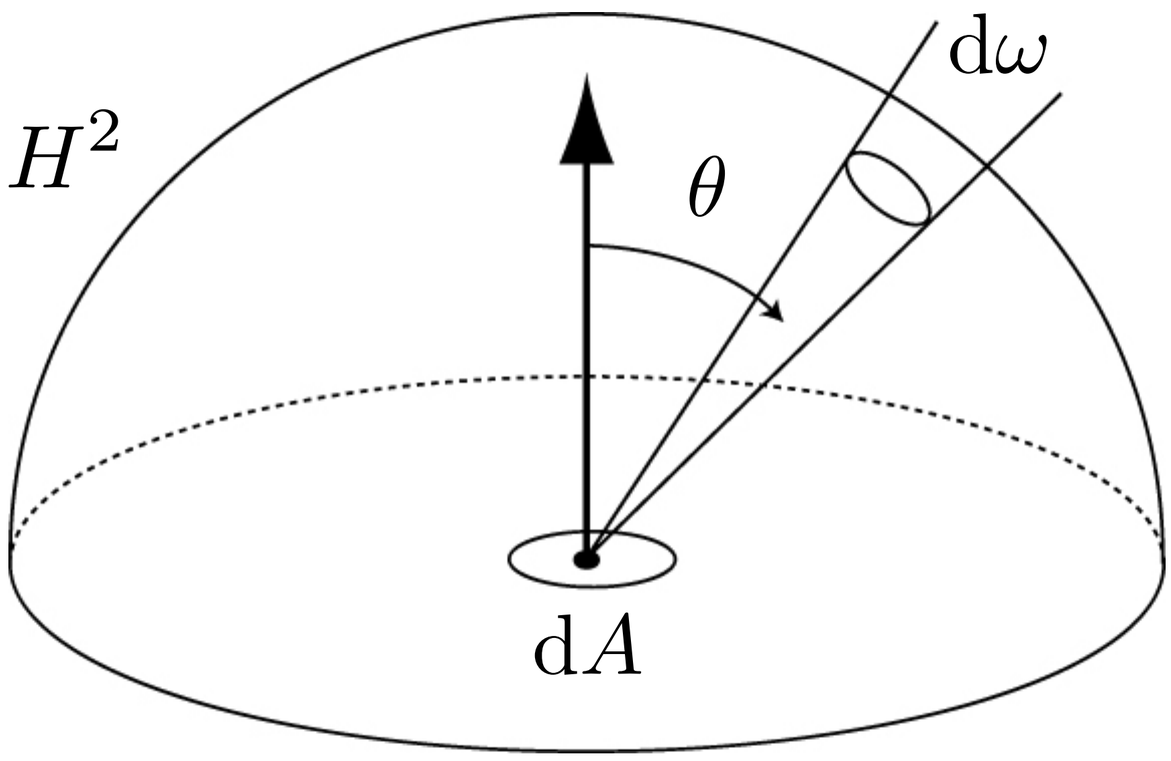
← **Contribution to irradiance from light arriving from direction ω**

$$E(\mathrm{p}) = \int_{H^2} L_i(\mathrm{p},\omega)\cos\theta\,\mathrm{d}\omega$$



Light meter

Hemisphere: H^2



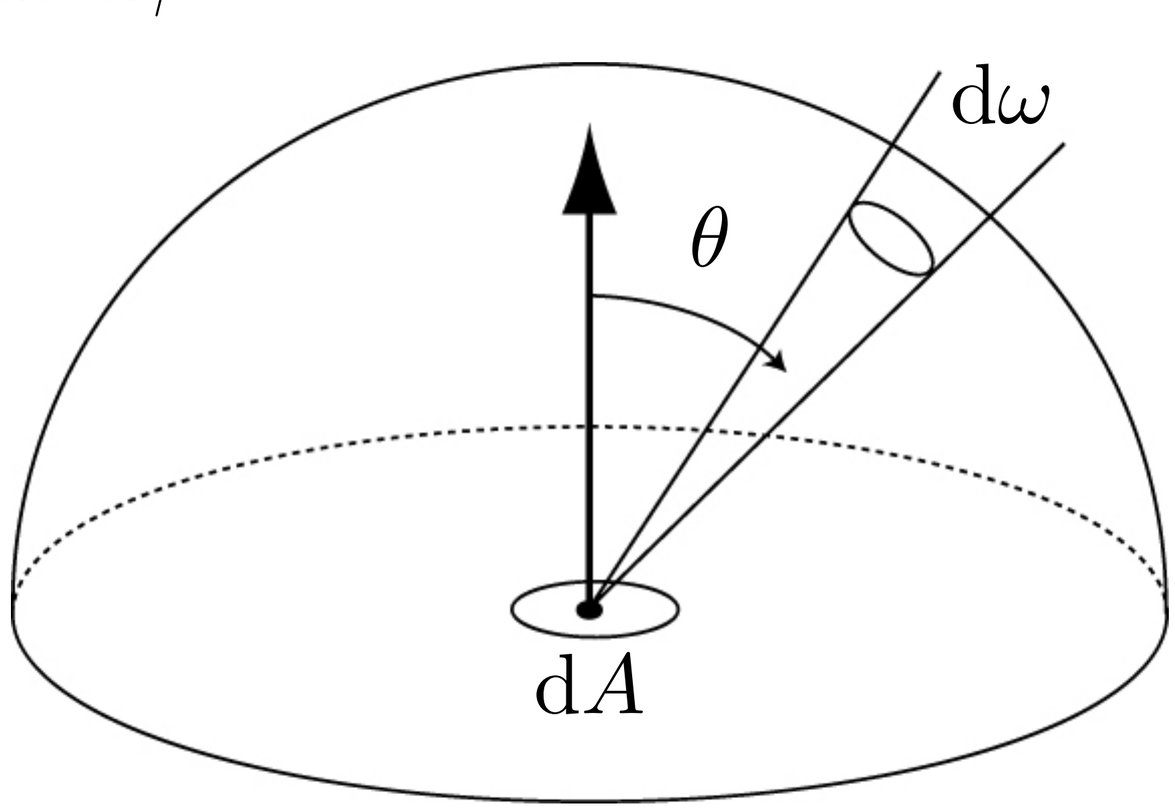
Irradiance from Uniform Hemispherical Light

$$E(\mathrm{p}) = \int_{H^2} L \cos\theta \, \mathrm{d}\omega$$

$$= L \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \cos\theta \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi$$

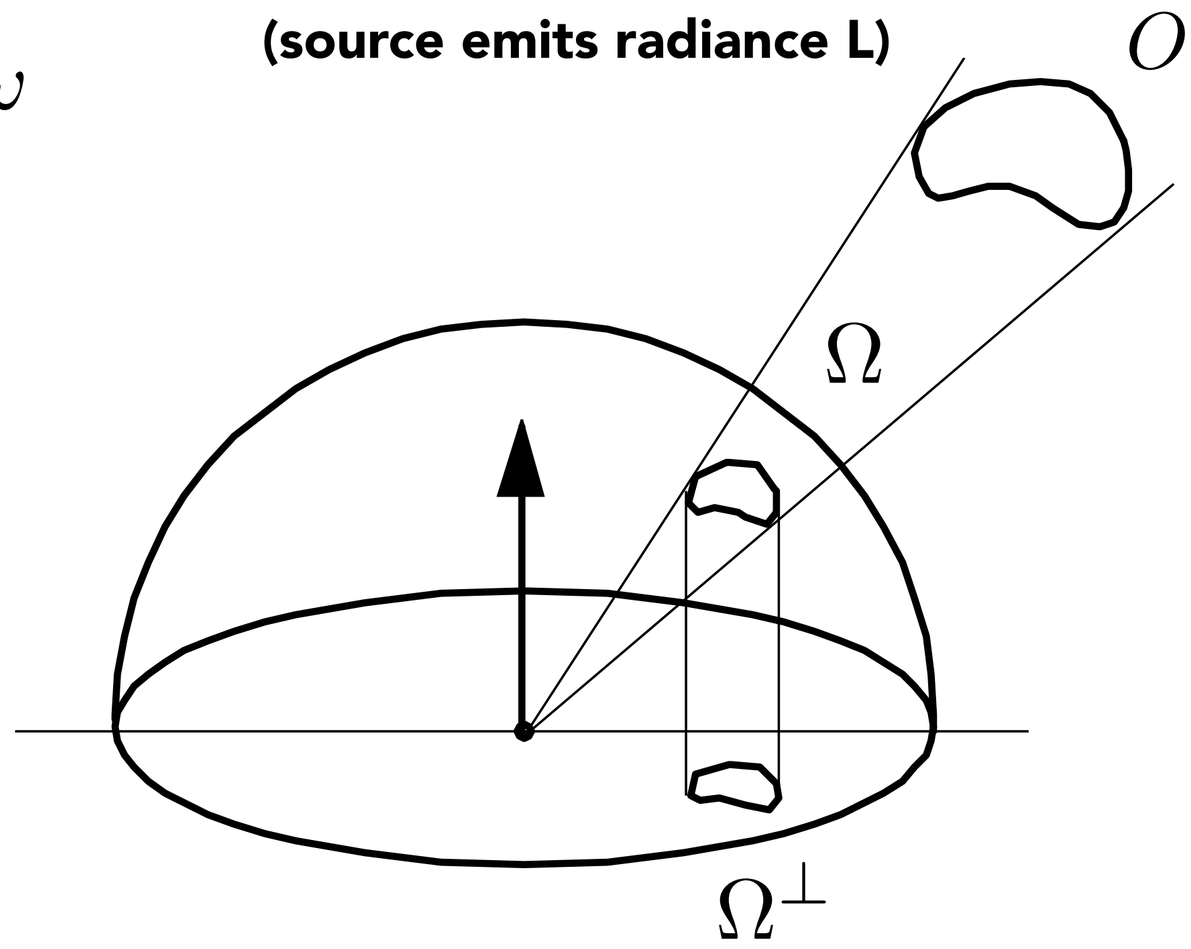
$$= L\,\pi$$

Note: integral of cosine over hemisphere is only 1/2 the area of the hemisphere.



Irradiance from a Uniform Area Source

$$
\begin{aligned}
E(\mathrm{p}) &= \int_{H^2} L(\mathrm{p},\omega)\ \cos\theta\,\mathrm{d}\omega \\
&= L \int_{\Omega} \cos\theta\,\mathrm{d}\omega \\
&= L\Omega^{\perp}
\end{aligned}
$$



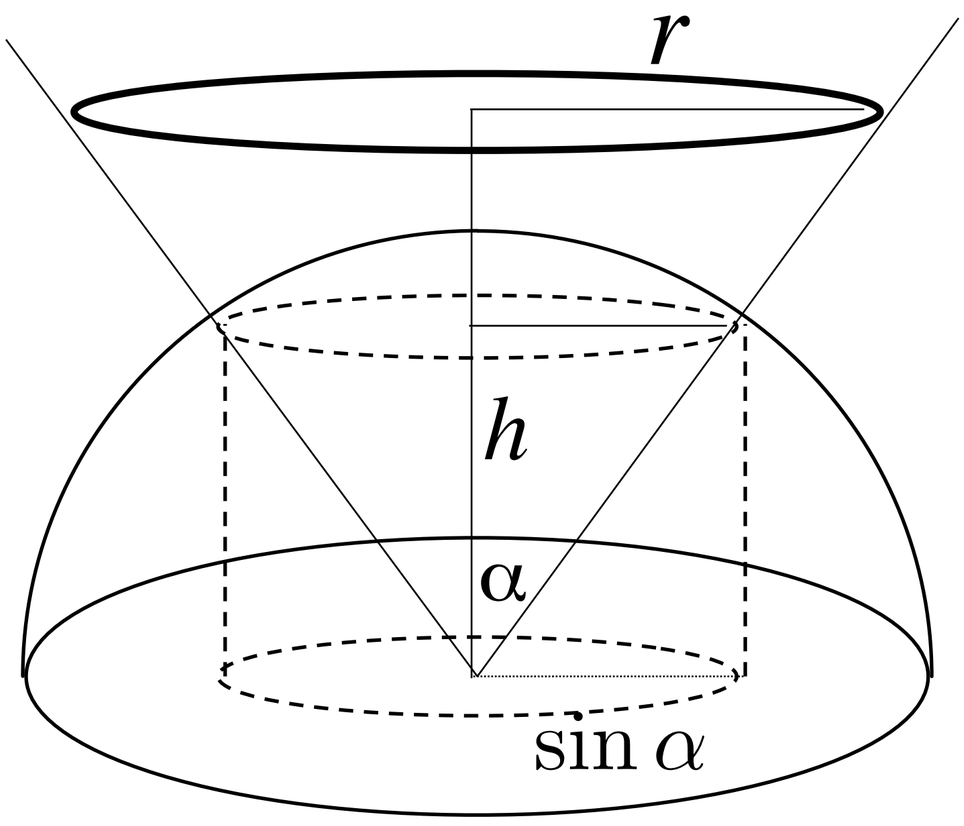
Projected solid angle:

- **Cosine-weighted solid angle**
- **Area of object *O* projected onto unit sphere, then projected onto plane**

$$\mathrm{d}\omega^{\perp} = |\cos\theta|\,\mathrm{d}\omega$$

Uniform Disk Source Overhead

Geometric Derivation



$$\Omega^{\perp} = \pi \sin^2 \alpha$$

Algebraic Derivation

$$\Omega^{\perp} = \int_0^{2\pi} \int_0^{\alpha} \cos\theta \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi$$

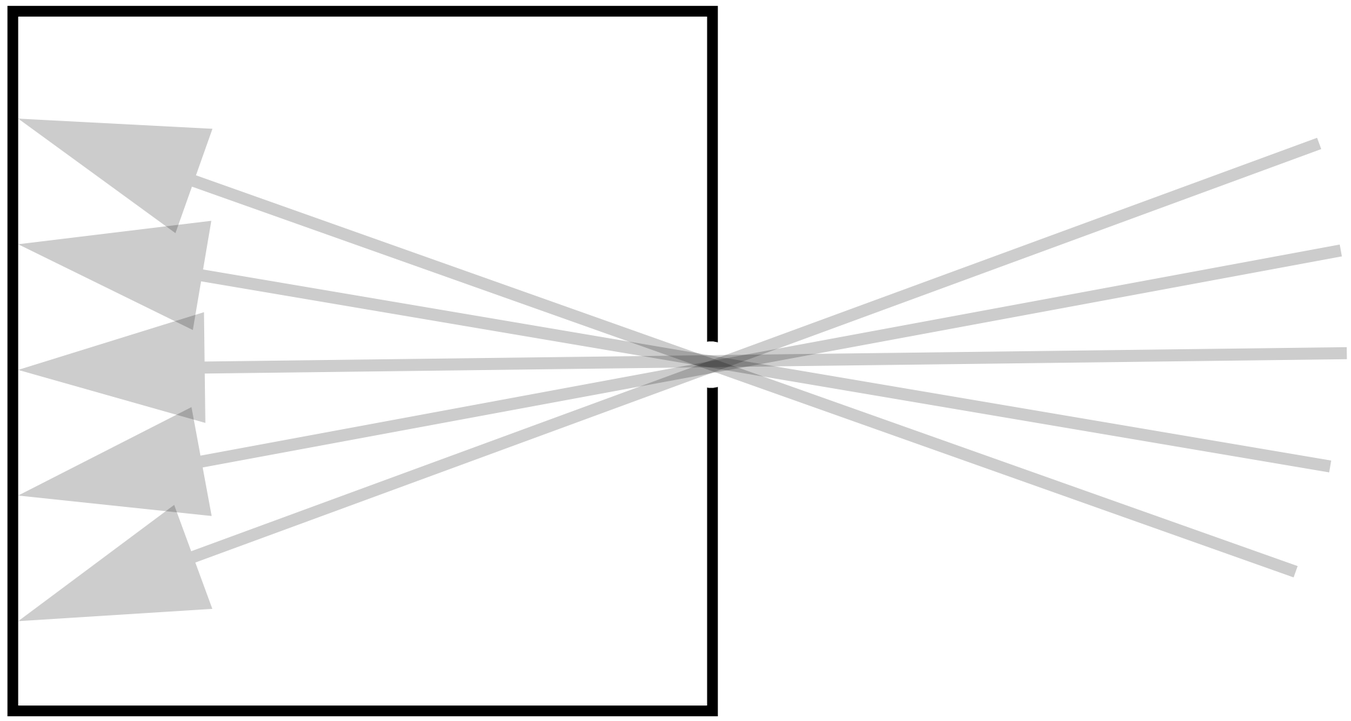
$$= 2\pi \frac{\sin^2 \theta}{2} \Bigg|_0^{\alpha}$$

$$= \pi \sin^2 \alpha$$

Measuring Radiance

A Pinhole Camera Samples Radiance

Photograph pixels measure radiance for rays passing through pinhole in different directions

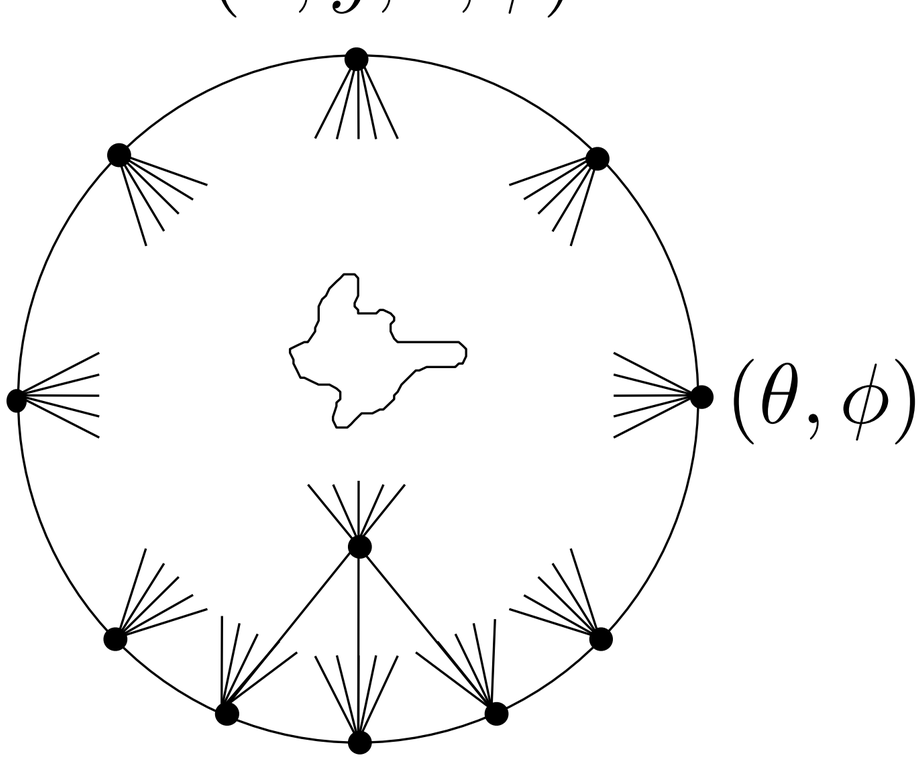


Spherical Gantry ⇛ 4D Light Field

Take photographs of an object from all points on an enclosing sphere

Captures all light leaving an object – like a hologram

$$L(x, y, \theta, \phi)$$

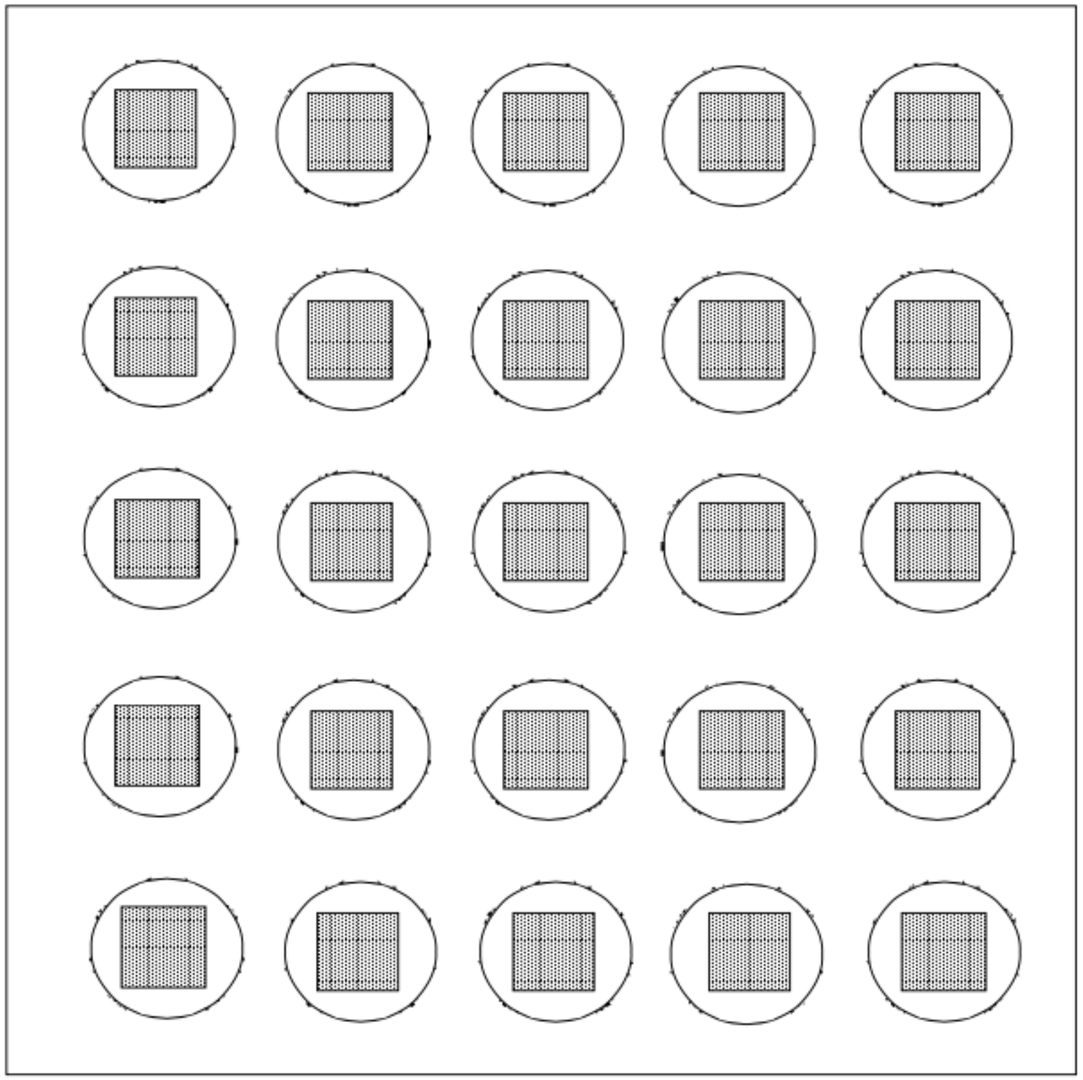




Multi-Camera Array ⇒ 4D Light Field



Two-Plane Light Field



2D Array of Cameras



2D Array of Images

Radiometry & Photometry

Terms & Units

Radiometric & Photometric Terms & Units

Physics		Radiometry	Units	Photometry	Units
Energy	Q	Radiant Energy	Joules (W·sec)	Luminous Energy	Lumen·sec
Flux (Power)	Φ	Radiant Power	W	Luminous Power	Lumen (Candela sr)
Angular Flux Density	I	Radiant Intensity	W/sr	Luminous Intensity	Candela (Lumen/sr)
Spatial Flux Density	E	Irradiance (in) Radiosity (out)	W/m²	Illuminance (in) Luminosity (out)	Lux (Lumen/m²)
Spatio-Angular Flux Density	L	Radiance	W/m²/sr	Luminance	Nit (Candela/m²)

"Thus one nit is one lux per steradian is one candela per square meter is one lumen per square meter per steradian. Got it?" — *James Kajiya*

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